Weather Impact on the Operations of Utrecht Centraal Station

Project Group - 20

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Contribution Statement

Author 1:

- Writing Chapter 1.3, Chapter 2, Chapter 3.1.1, Chapter 3.3.1;
- Coding Chapter 3.1.1 and Chapter 3.3.1

Author 2: Chapter 1.1, 1.2, 3.1.3

Author 3: Chapter 3.3.3, Coding 3.3.3, Discussion, Conclusion

Author 4:

- Writing: Chapter 1.2, Chapter 3.2, Chapter 3.3.2, Discussion & Limitations, Discussion & Recommendations
- Coding: files in total = train_analysis.py, train_analysis.ipynb, data_import.py, train_comparison.py, initialized weather_comparison.py (Chapter 3.2, 3.3)
- Additional: Quality Check, README.txt, Data Loading, Assumptions, Troubleshooting peers code

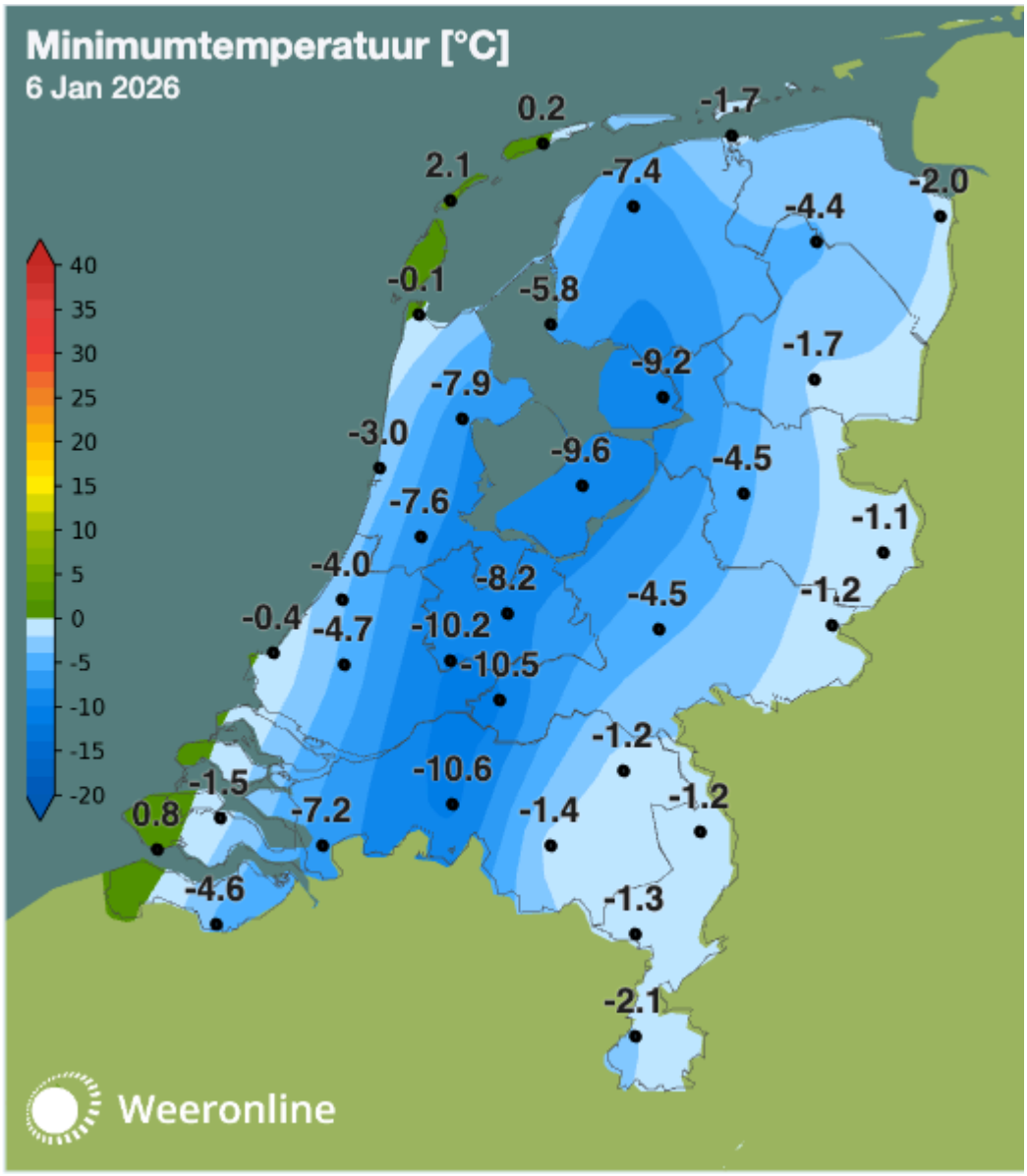
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1. Introduction

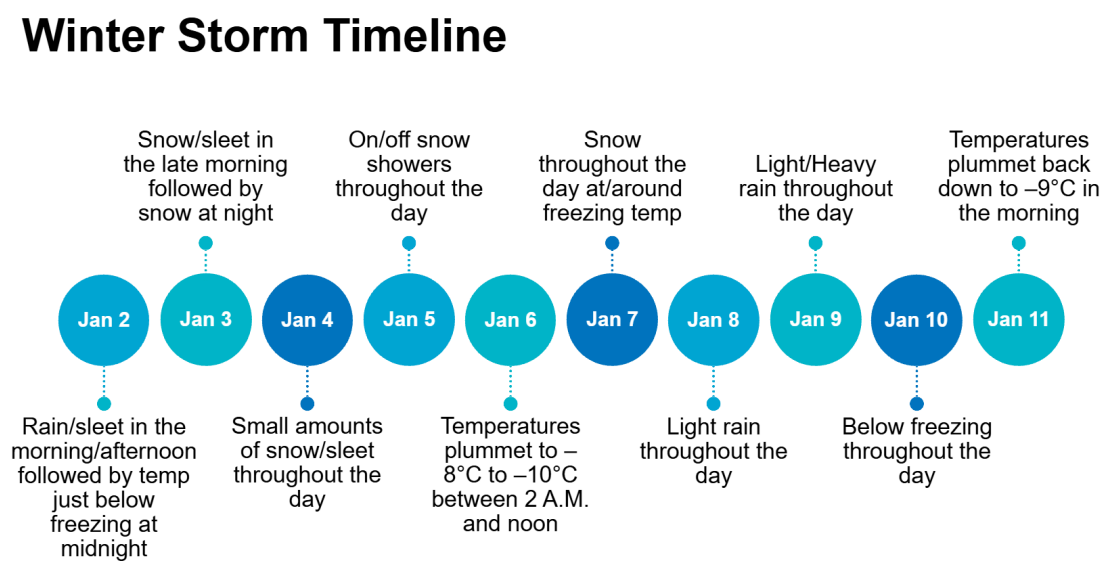
Railway operations are highly dependent on weather conditions, as severe weather acting in both ways can affect the quality of operations. Severe weather conditions can affect railway infrastructure as well as the operation of train services. Winter conditions, such as freezing temperatures and snowfall, can create challenges for operations resulting in delays and cancellations. This research aims to investigate the impact of a severy (winter) weather event on railway operations at Utrecht Centraal.

1.1 Context

In early January of 2026, a winter storm hit The Netherlands with below freezing temperatures and snow up to around 20 cm. The most affected area, as seen in the figure below [7], was the area around Utrecht.



As seen in the timeline below with weather data gathered from KNMI (2026), the storm/cold weather started on January 2 and lasted until about January 11, although the worst of it occurred January 6, with temperatures going as low as -10°C in the early morning.



The main aspect of the winter weather that causes trouble for rail transit is the freezing temperatures. When temperatures get as low as they did on January 6, it can cause ice to form on the tracks and catenary wires above the tracks. Ice on the tracks creates issues for the rail switches as they can literally freeze in place. Areas with colder climates that experience these issues more often have ways to mitigate these effects, mainly through heating the rails [8]. Additionally, ice on the catenary wires can interfere with electricity transfer between the train and the wires.

The disruption data provided by Rijden de Treinen [2] shows that these weather conditions caused disruptions to railway infrastructure in and around Utrecht Centraal. This is particularly relevant given that Utrecht Centraal is the largest railway station in the Netherlands in terms of passenger throughput [6]. Located in one of the regions most severely affected by the winter storm, the station also has an extensive railway infrastructure, including approximately 70 switches [4, 5]. The combination of its high operational intensity, extensive infrastructure and exposure to severe winter conditions makes Utrecht Centraal an interesting case for investigating the impact of extreme winter weather on railway operations.

1.2 Problem statement

Utrecht Centraal's high train fequency and extensive railway infrastructure make it vulnerable to disruptions, amongst others caused by severe winter weather. During the snowstorm in January 2026, freezing temperatures as well as snowfall affected the infrastructure of the railway in and around the station. This potentially caused switch and power supply malfunctions. These issues can disrupt train operations, resulting in increased delays, cancellations and reduced punctuality.

Although the effects of severe winter weather on the infrastructure are understood, being a decrease in operations, the extent to which the snowstorm actually affected train operations at Utrecht Centraal has not been quantified. It is important to establish how performance during a heavy disruption differs from performance under normal

conditions, as this can provide insight into the impact of severe winter weather on railway operations.

1.3 Research Question & Objectives

Derived from the problem statement the following research question was formulated: *How did the snowstorm event from the 2nd of January to the 11th of January affect the duration of train delays, the number of cancellations, and punctuality at Utrecht Centraal, compared with a typical period under normal weather conditions?*

Based on the above research question the following six objectives have been set:

1. Establish a baseline for the weather conditions at Utrecht Centraal from the 2nd of January to the 11th of January, in terms of temperature, precipitation, dew point, windspeed and snow occurrence.
2. Establish a baseline of NS operations and performance at Utrecht Centraal from the 2nd of January to the 11th of January, in terms of delay duration, number of cancellations, and punctuality.
3. Visualize and describe the weather patterns at Utrecht Centraal during the snowstorm event, using temperature, precipitation, dew point, windspeed and snow occurrence.
4. Visualize and describe NS operations and performance at Utrecht Centraal during the snowstorm event, using delay duration, number of cancellations, and punctuality.
5. Compare the baseline of NS with the performance of NS at Utrecht Centraal during the snowstorm event in order to determine the impact of the snowstorm event on delay duration, number of cancellations, and punctuality.
6. Compare the weather scenarios with the NS train scenarios in order to determine the relationship between the weather indicators and the performance indicators.

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2. Methodology

This chapter describes the data collection and analysis methods, including data imports (paragraph 2.1), data used and how it was proccessed (paragraph 2.2), and the data pipeline (paragraph 2.3).

2.1 Imports

The data files are imported using the code below.

In [19]:
```
import importlib
import background.train_analysis
import background.weather
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt

importlib.reload(background.train_analysis )
importlib.reload(background.weather)

from background.train_analysis import (
    load_train_data,
    filter_cancellation_days,
    analyse_delays,
    analyse_delays_per_hour,
    analyse_punctuality,
    analyse_delay_absorption,
    analyse_correlations,
    cancellation_chart
)
from background.weather import(
    load_weather_data,
    set_format,
    plot_temp_and_snow_occur,
    plot_temp_and_sleet_occur,
    plot_temp_vs_dew_point,
    plot_temp_and_precipitation,
    plot_temp_and_wind_speed,
    plot_baseline_weather,
    plot_event_vs_baseline,
    plot_baseline_year_by_year,
    plot_year_by_year,
    baseline_daily_statistics,
    event_vs_baseline_statistics,
    baseline_yearly_statistics,
    yearly_statistics_table
)

df_weather = load_weather_data()

train = load_train_data()
train = filter_cancellation_days(train, cancellation_percentile=90)
```

2.2 Data Used

The datasets used as input in this project were acquired from KNMI [1] and Rijden de Treinen [2]. The weather dataset specifically was from the weather station De Bilt, which was chosen as the representative proxy of weather patterns around Utrecht Centraal due to its relatively close distance to the station.

2.2.1 Weather dataset

The weather dataset contained detailed hourly records of weather records per day at De Bilt (weather station 260), 4 km northeast of Utrecht Centraal, including but not limited to:

- Temperature
- Dew point
- Precipitation amount and duration
- Snow occurence (weather code)
- Wind speed

Initially, the raw data was loaded from the CSV file, and unnecessary formatting characters and whitespace were removed from the column names using Claude Sonnet 5.5 Medium, making sure to keep missing data traceable.

Next, the date and time information was processed to create a datetime index for each hourly observation. Since KNMI records its hourly observations using hour-ending UTC timestamps (1–24), one hour was subtracted to represent the actual start of each observation period. The weather measurements were then converted into standard units to make them easier to interpret. Temperature and dew point were converted from tenths of degrees Celsius to degrees Celsius, precipitation from tenths of millimetres to millimetres, and wind speed from tenths of metres per second to kilometres per hour. Precipitation values recorded as -1, representing trace amounts, were treated as zero. A binary snow indicator was also created to identify hours with recorded snowfall.

The dataset was subsequently sorted chronologically and restricted to the study period 2–11 January for each year from 2021 to 2026. This allowed the weather conditions in January 2026 to be compared with the same calendar period across the five preceding years (2021–2025), which was deemed a sufficiently large sample for the purposes of this research. Each observation was also assigned a day index, indicating its position within the selected 10-day period.

Finally, several weather indicators were calculated at both daily and yearly levels. Daily indicators included average temperature, average dew point, total precipitation, average wind speed and the number of hours with snowfall. Yearly indicators summarised the entire 10-day period for each year and included average and minimum temperatures, total precipitation, average wind speed, and snowfall hours.

These calculations produced two datasets: one containing daily weather indicators for comparing corresponding days across years, and another containing overall weather indicators for comparing the full 10-day periods. The 2021–2025 observations were then separated to serve as the historical baseline against which the weather conditions (snowstorm event) in January 2026 could be evaluated.

Note that two CSV weather files were created in the analysis process, one with only the data from 2026 (weather_selected.csv) and another with the full dataset from 2021 to 2026 (weather_total_cleaned.csv). This was done out of practicality and purely served the purpose of maintaing overview and debugging. Besides the difference in filtered years, the CSV files are identical.

2.2.2 Train dataset

The train dataset contained detailed records of train arrivals, departures, delays and cancellations, including:

- Service type (e.g. Sprinter, InterCity)
- Cancellation type (completely, partly)
- Maximum delay for a certain service
- Arrival delay
- Departure delay
- Arrival or departure cancelled

The dataset was first restricted to the study period of 15 December 2025 to 30 April 2026. Unnecessary columns were removed, and the remaining columns were renamed to improve clarity and consistency. The analysis was limited to Intercity and Sprinter services, excluding all other train types.

Next, scheduled arrival and departure times were converted into a consistent hour-and-minute format. Actual arrival and departure times were then calculated by adding the recorded delays (in minutes) to their respective scheduled times. Based on these actual times, the dataset was restricted to train stops where both arrival and departure occurred between 05:00 and 22:59. The data was subsequently sorted by date and actual arrival time.

Several additional variables were created to support the analysis. Binary indicators were introduced to identify whether a train stop experienced an arrival or departure delay, with separate indicators identifying delays of at least three minutes. A cancellation indicator was also created, classifying a stop as cancelled if either its arrival or departure was cancelled. Missing cancellation values were treated as not cancelled. Additionally, a weekly variable was generated to group observations by calendar week.

Finally, to prevent days with unusually high cancellation activity from disproportionately influencing the baseline analysis, a filtering procedure was applied. Daily cancellation totals were calculated, and the 90th percentile was used as the exclusion threshold. Days with cancellation counts strictly above this threshold were removed, while days at or below the threshold were retained. This filtering step was optional and could be disabled to retain all days.

2.3 Data Pipeline

The data analysis was split into two teams: Team Weather and Team Train. These two teams worked in parallel for steps 1 to 8, coming together to do step 9.

The complete data pipeline consisted of the following general steps:

1. Import the datasets

2. Clean, filter, and process the data based on assumptions and the research scope
3. Calculate the relevant indicators
4. Establish the two baselines
5. Identify and analyse performance during the snowstorm period.
6. Compare snowstorm performance with the baselines.
7. Generate visualizations and findings with respect to the relevant indicators
8. Combine visualizations and findings across teams to gain insight into the relationship between indicators.

The following steps were specifically taken by Team Weather:

1. Import the weather dataset from KNMI
2. Process dates and hours and convert KNMI scaled values to standard units
3. Clean and filter the data based on on assumptions, and the Utrecht Centraal study period.
4. Calculate weather indicators, including temperature, dew point, percipitation, windspeed, and snow occurence.
5. Establish baseline performance of typical weather patterns.
6. Identify and analyse weather patterns during the snowstorm period.
7. Compare snowstorm performance with the baseline.
8. Generate visualizations and findings for the operational impact of the snowstorm.

The following steps were specifically taken by Team Train:

1. Import the train dataset from Rijden de Treinen
2. Clean and filter the data based on assumptions, relevant services, and the Utrecht Centraal study period.
3. Process arrival and departure delays and identify cancelled services.
4. Calculate operational performance indicators, including delay duration, cancellations, and punctuality.
5. Establish baseline performance under typical weather conditions.
6. Identify and analyse operations during the snowstorm period.
7. Compare snowstorm performance with the baseline.
8. Generate visualizations and findings for characteristics of the snowstorm.

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3. Results

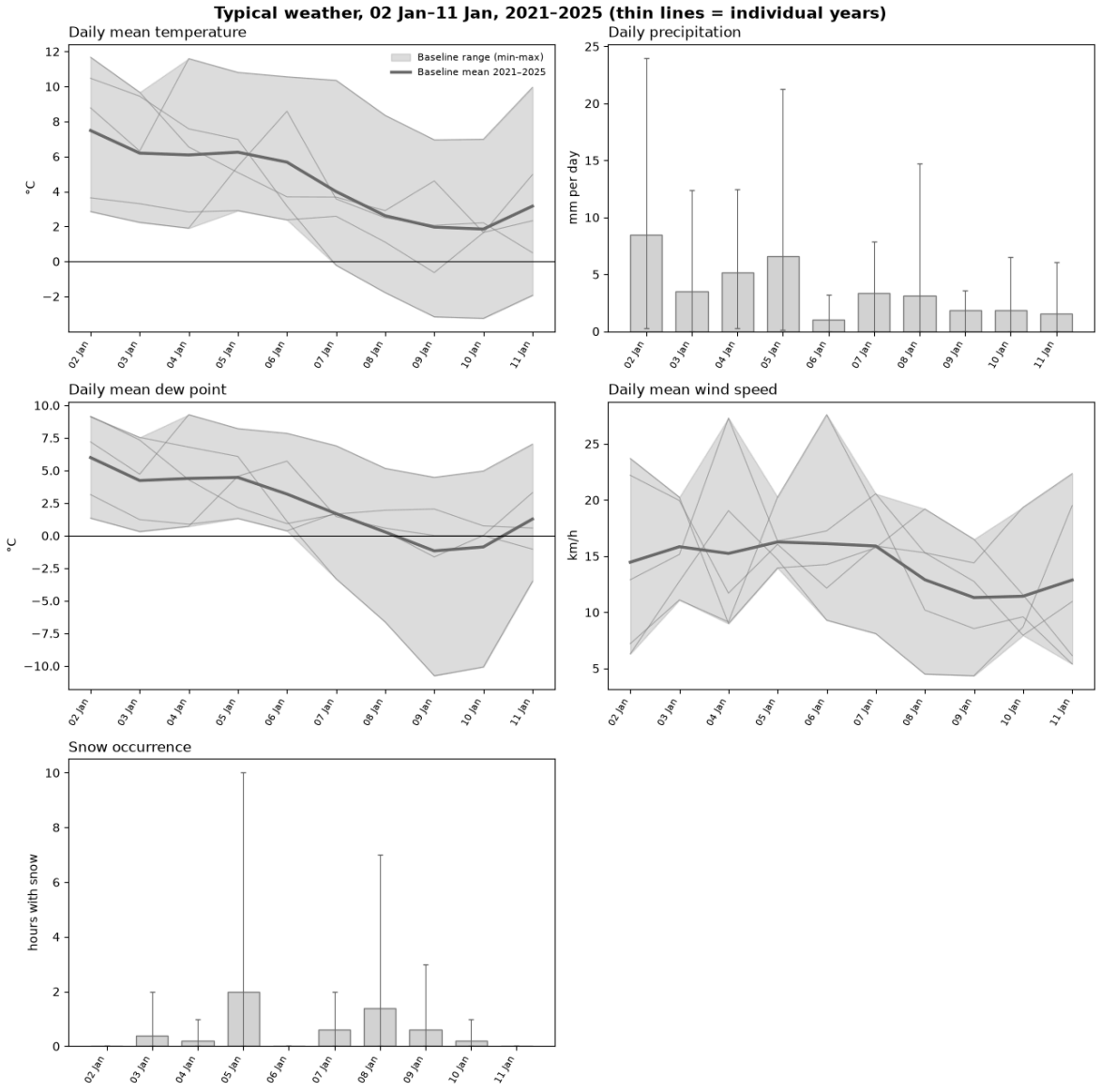
Section 3.1 describes the weather analysis, including the construction of a weather baseline and the snowstorm analysis. The following Section (3.2) establishes a baseline performance for the train operations at Utrecht Centraal, including the performance of the train operations during the snowstorm. Finally, Section 3.3 addresses the comparison analysis, where the typical scenarios are compared with the adverse scenarios.

3.1 Weather analysis

3.1.1 Weather baseline

To construct a baseline for the weather patterns around Utrecht Centraal, five weather indicators were used: temperature, dew point, precipitation, wind speed and snow occurence. Based on these weather indicators, the weather patterns for the years 2021 to 2025. It must be stated that these years did not contain any adverse weather patterns, and thus were deemed appropriated for establishing the baseline weather patterns around Utrecht Centraal. The weather characteristics with respect to these five indicators are shown in the graphs below.

In [20]:
```
fig_baseline_daily = plot_baseline_weather()
```



Based on the above graphs and statistics, the following conclusions have been drawn:

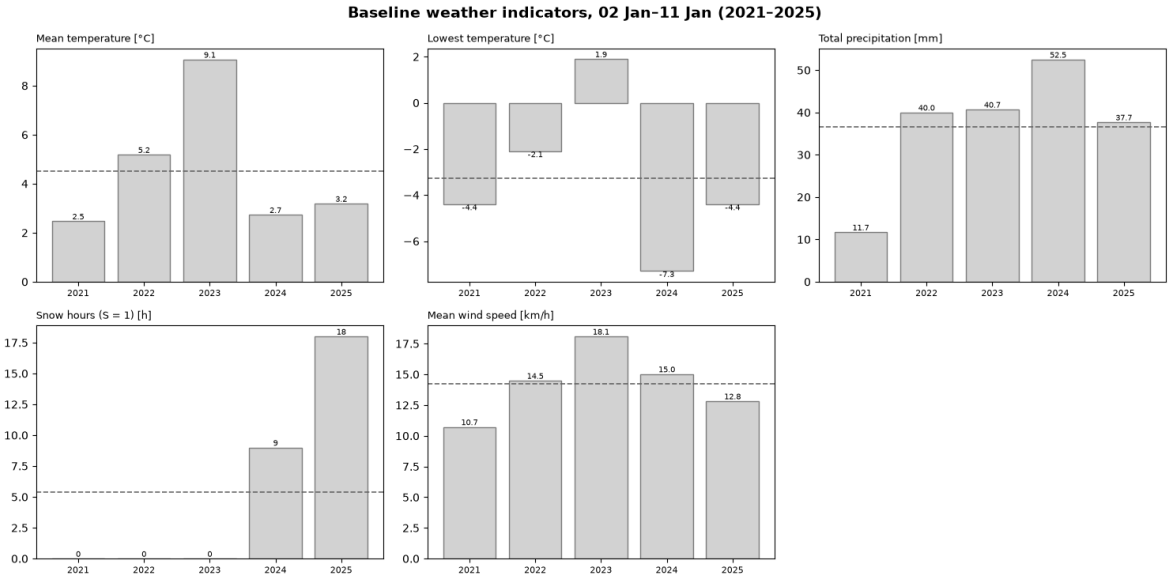
- Daily mean temperature consistently stayed above freezing, averaging between 1.8°C and 7.5°C. However, the relatively wide range for typical weather patterns

around Utrecht Centraal suggests significant temperature volatility during this period.

- Compared to daily mean temperature, the daily dew point displayed a narrower typical range, averaging between -1.2°C and 6.0°C. This suggests that dew point is a more reliable indicator of standard weather patterns at Utrecht Centraal. Still, a notable exception occurred in one specific year, where the mean dropped below 0°C on January 9th and 10th.
- Both daily mean temperature and dew point trended downward from January 2nd until January 9th/10th, after which they began to rise.
- Most days exhibited a wide range of typical precipitation levels, with noticeable exceptions on January 6th and 9th. Overall, this indicator showed a general downward trend across the 10-day period.
- Daily mean wind speed seemed to display the most stable average of the study period, fluctuating only between 11.3 km/h and 16.3 km/h, despite maintaining a fairly broad typical weather zone.
- Snow occurrence remained generally low, averaging under two hours per day across the 10-day period. However, January 5th and 8th stood out with relatively large typical ranges.

Besides the daily weather patterns, yearly weather patterns were also analysed. Similarly to the above graph, yearly data was plotted in the graph below with respect to the five weather indicators.

In [21]:
```
fig_baseline_yearly = plot_baseline_year_by_year()
```



Based on the graphs and statistical analysis presented above, the following conclusions have been drawn:

- The baseline period recorded a mean temperature of 4.5°C, with annual averages ranging from 2.5°C to 9.1°C. Notably, 2023 experienced a considerably higher average temperature than the other four years, indicating that it was an unusually warm year within the baseline period.

- Mean minimum temperature across the baseline period was -3.7°C, with values ranging from -4.4°C to 1.9°C. Notably, four out of the five years recorded average minimum temperatures below 0°C, with 2023 being the only exception. This further supports the observation that 2023 was a relatively warmer year compared to the rest of the baseline period.
- Total annual precipitation amounted to 36.5 mm, with values ranging from 11.7 mm to 52.5 mm. However, 2021 recorded a considerably lower value, suggesting that it was an exceptionally dry year. Excluding 2021, precipitation levels remained relatively stable throughout the baseline period.
- Snow occurrence demonstrated a notable upward trend over the baseline period. No snow was recorded during the first three years, while occurrence increased to 9 in 2024 and doubled to 18 in 2025. This resulted in an average of 5.4 snow hours per year.
- Mean wind speed across the baseline period was 14.2 km/h, with annual averages ranging from 10.7 to 18.1 km/h. The relatively limited variation around the overall average suggests that wind speed remained comparatively stable and consistent throughout the period.

3.1.2 Baseline weather conclusions

The 10-day analysis at Utrecht Centraal provides an overview of short-term weather variability. Daily mean temperatures remained above freezing, although both temperature and dew point followed a similar downward trend until January 9th and 10th, before beginning to rise again. Dew point showed a narrower typical range than temperature, suggesting more consistent daily patterns. Precipitation generally followed a downward trend, while wind speed remained the most stable indicator throughout the observation period. Snow occurrence was limited overall, averaging fewer than two hours per day, although January 5th and 8th showed greater variability.

Looking at the broader annual baseline, weather conditions were generally relatively stable, with some notable variations between years. Temperatures were generally moderate, although 2023 stood out as a considerably warmer year than the rest of the baseline period. Precipitation levels also remained relatively consistent, with the exception of 2021, which was notably drier. Snow occurrence increased towards the end of the baseline period, while wind speeds remained relatively stable throughout.

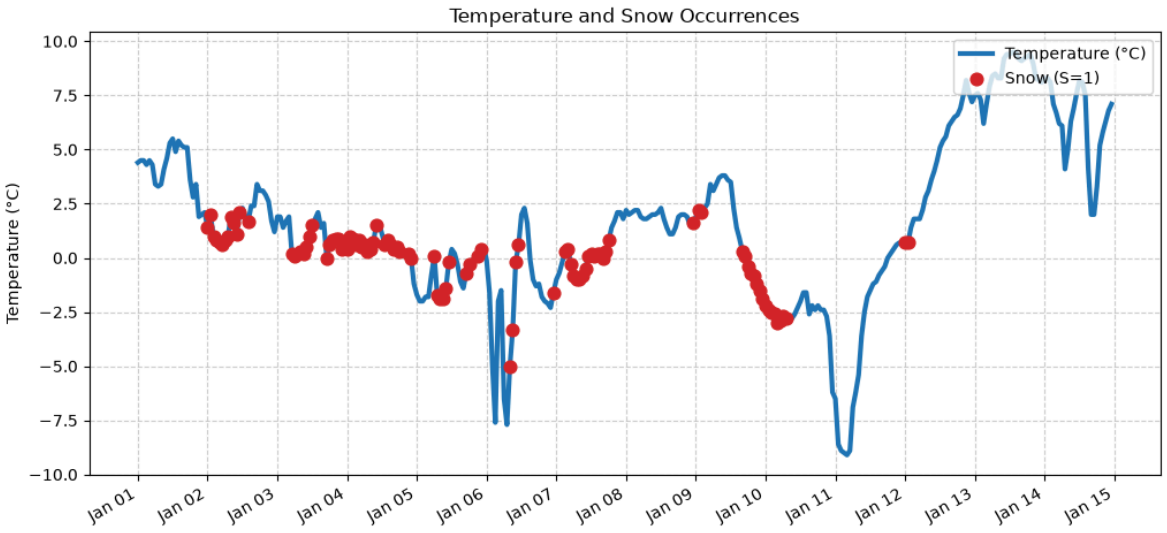
Taken together, the findings suggest that while the annual baseline was characterised by relatively consistent precipitation and wind conditions, temperature, snow occurrence and short-term weather patterns exhibited greater variability. The daily analysis further highlights how weather conditions can fluctuate considerably over a relatively short period, even within an otherwise relatively stable broader baseline.

3.1.3 Snowstorm analysis

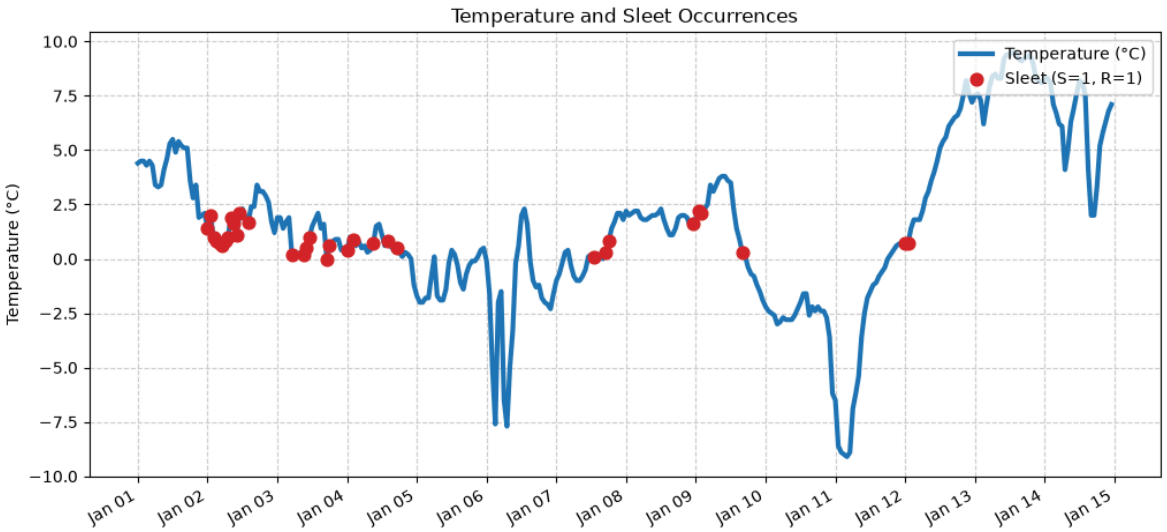
The weather data was retrieved from KNMI at their local weather station in De Bilt. The graphs shown below help illustrate the timeline of the winter storm and what occurred

between January 2-11. The following two graphs illustrate the relationship between precipitation and temperature during the storm. The first graph plots the temperature with snow occurrences, which in the data set are marked as a binary variable; "1" indicating that snow occurred during the hour and "0" indicating no snow occurring during the hour. The second graph plots temperature against hours when both snow and rain occurred, which in combination, forms sleet. In the same way as the snow, rain is also used as a binary variable in the dataset.

In [22]:
```
plot_temp_and_snow_occur(df_weather)
```

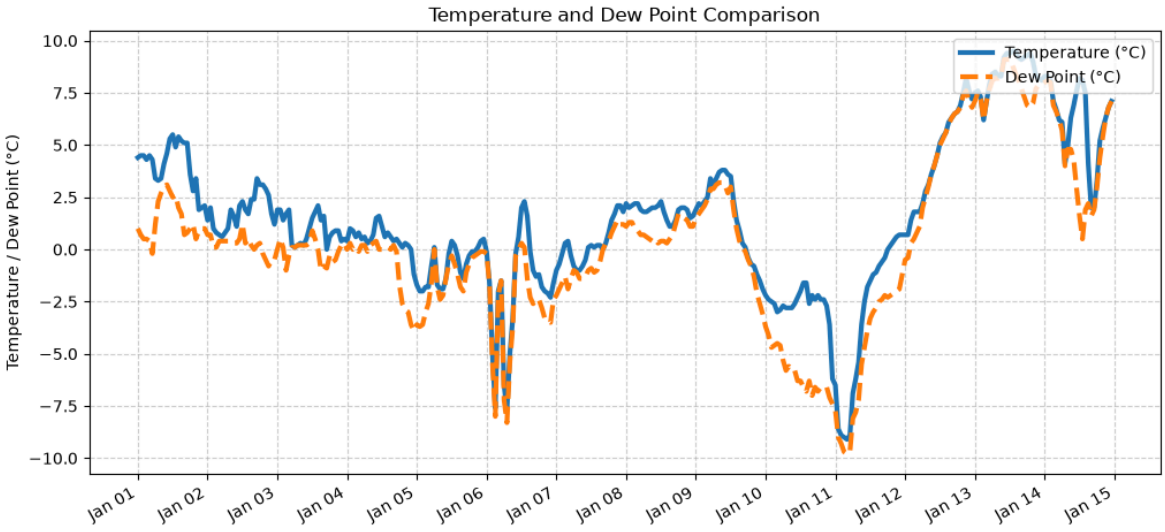


In [23]:
```
plot_temp_and_sleet_occur(df_weather)
```



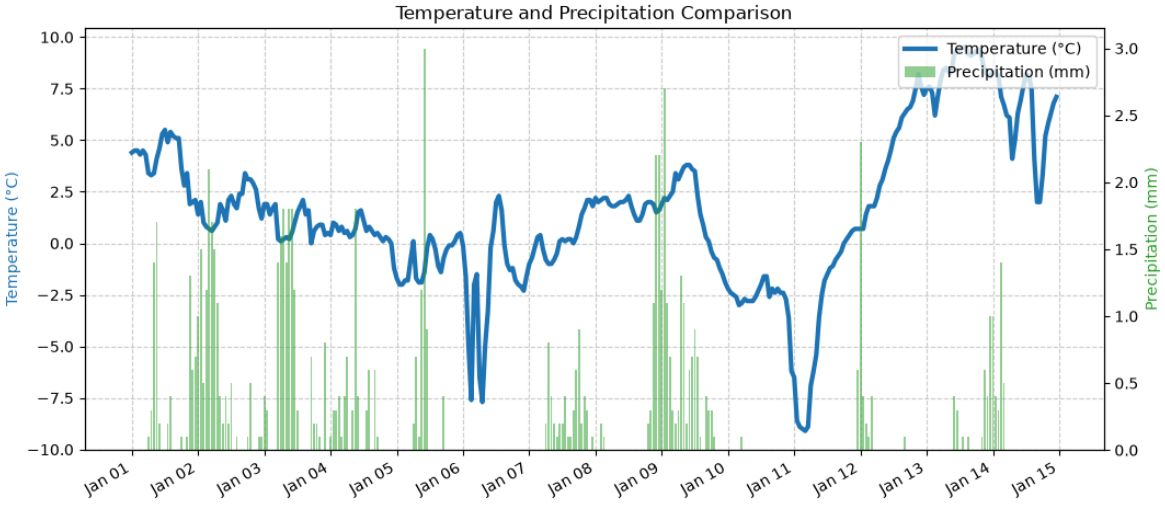
In the following graph, the dew point has been plotted against the temperature. The dew point is a representation of the temperature at which water will start to condense onto surfaces. This graph shows that the morning of January 6, the temperature was as low as the dew point when the temperatures were below freezing. This indicates that frost formed onto surfaces during this period of time. This is an indicator that ice formation occurred during the morning of January 6.

In [24]:
```
plot_temp_vs_dew_point(df_weather)
```



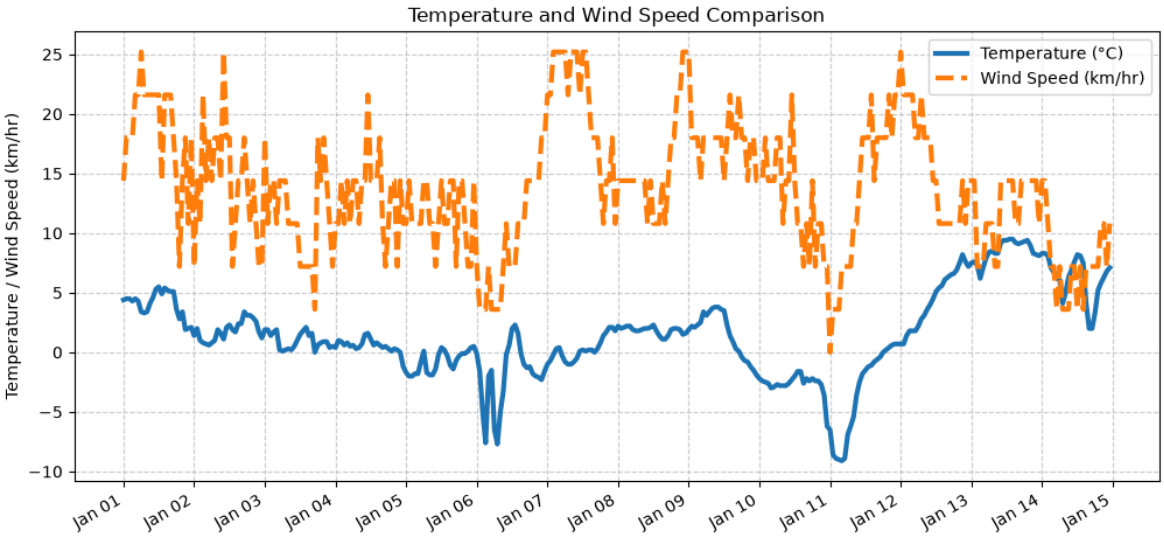
The following graph plots temperature related to precipitation. As seen in the graph, the most precipitation occurred during January 5 as well as from January 9-10. The precipitation on January 5 helped cause the scenario of ice formation that occurred the morning of January 6.

In [25]:
```
plot_temp_and_precipitation(df_weather)
```



The following graph plots temperature with wind speed. As seen in the graph, wind speed did not have that large of an affect in the storm, as it was fairly calm when the temperatures went down. The only time wind speed could be a relevant factor is on January 7 and 8, when the wind speed went up to 25 km/hr, which is still a relatively gentle breeze, especially for The Netherlands.

In [26]:
```
plot_temp_and_wind_speed(df_weather)
```



3.1.4 Snowstorm conclusions

As seen by the various weather indicators and data provided by KNMI, snow/sleet occurred from January 2 to January 5. However, the temperature falling to about -8°C and the dew point matching the temperature during that time provided ample opportunity for freezing of the railway infrastructure to occur. In the following train analysis, the weather indicators will be paired against cancellations that occurred during the period of January 2-11.

3.2 Train analysis

The train data usage, cleaning and modification is described in section 2.2. For background, see /background/train_analysis.ipynb and background/train_analysis.py.

The train analysis creates a baseline for the comparison analysis; an initial data pipeline is created to identify the generic indicators and insights which can be retreived from the data.

The three indicators and justification used are as follows:

- Delay indicator:
  - Every arrival- and departure delay is flagged if the delay column > 0.
  - An additional binary indicator is more important, however, which is the arrival and departure delay being bigger or equal to three (3) minutes or 180 seconds. This is the delay value indicated in NS' performance agreements as communicated to passengers and stakeholders [3].
- Number of cancellations:
  - A train is cancelled if either arrival or departure is cancelled.
  - Cancellations can be per hour, day, week of day or the average over the analysis period.
- Punctuality performance:

- Punctuality of a train is the number of on-time events (on-time indicating delays being smaller than 180 seconds) divided by the total scheduled events.
- Punctuality can differ at arrival, departure and a total punctuality figure.
- Keeping non-cancelled trains in this dataset

The delay indicator entails how much and how severe a disruption is, where the number of cancellations complement this. Punctuality answers the question how often the services meet the performance standard. The delay measures the magnitude and distribution of operational disruption, whereas the punctuality measures the proportion of services which meet a predefined performance standard.

3.2.1 Delays and cancellations in numbers

In [27]:
```
delayed_trains = (len(train[(train['arrival_delayed'] == True) | (train['departu
delayed_operations = len(train[(train['arrival_delayed_3'] == True) | (train['de
total_cancellations = len(train[train['cancelled'] == True])
total_cancellations_perct = total_cancellations / len(train) * 100

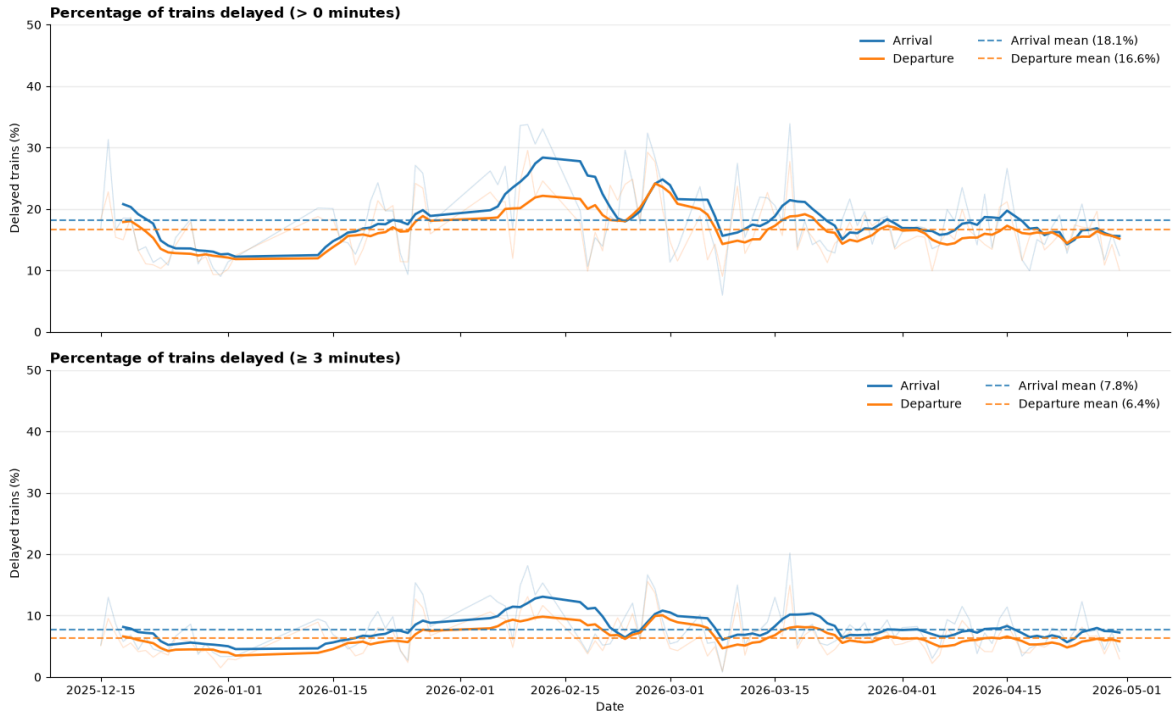
print(f"{delayed_trains:.1f}% of trains were delayed.")
print(f"{delayed_operations:.1f}% of train operations were delayed by more than
print(f"{total_cancellations_perct:.1f}% of trains were completely cancelled.")

delay = analyse_delays(train)
```

```
24.8% of trains were delayed.
9.6% of train operations were delayed by more than 3 minutes (actual delay classi
fication).
7.5% of trains were completely cancelled.
DELAYS AND CANCELLATIONS
Number of complete cancellations: 5707
    24.8% of trains were delayed.
    9.6% of train operations were delayed by at least 3 minutes.
    7.5% of trains were cancelled.

RELATIVE PERFORMANCE
    Delay-minutes per vehicle: 1.19 minutes
    Total delay-minutes per day: 822.1 minutes
    Cancellations per 100 vehicles: 7.49
    Cancellations per day: 51.88
```

Figure 3.2.1: Daily Train Delays Over Time



Across the full period, 25% of the trains experienced a delay greater than zero minutes, while around 10% were delayed by at least three minutes. This indicates that minor delays were relatively common. On average, each vehicle is delayed by 1.19 minutes per vehicle, while the total delay-minutes per day accumualted to 845.5 minutes in the baseline. The cancellations per 100 vehicles in the baseline equals 7.5 vehicles per 100. These results show that operational disruptions involve both delays and cancellations.

The figure (Fig. 3.2.1) shows the daily percentage of delayed trains in the time-period, for delays above 0 minutes and delays of 3 minutes or more. Thin lines show daily values and bold lines show the smoothed rolling trend. Arrivals are delayed more often than departures (mean 18.1% vs 16.6% for 0 minutes and 7.8% vs 6.4% for above 3 minutes), an early indicator that part of the arrival delay is absorbed during the stop.

Delays vary over time; the lowest levels occur around the year turnover, after implementation of the new timetable, and are followed by a sustained peak from February to end of February. Here, arrival delays reach around 28% and 13% (>0 min), from about 12% and 5% in December (>= 3 min). The thin lines indicate that delays fluctuate strongly, so the trend is the better indicator along the mean values.

3.2.2 Delays per hour

Next, we analyse the average delays, frequency of delays and cacellation rate per hour of operation over the dataset. As indicated earlier, operation is defined as the period from 0500 until 2300 (h= [5,22]).

In [28]:
```
delay_per_hour = analyse_delays_per_hour(train)
```

Figure 3.2.2: Average delay by hour

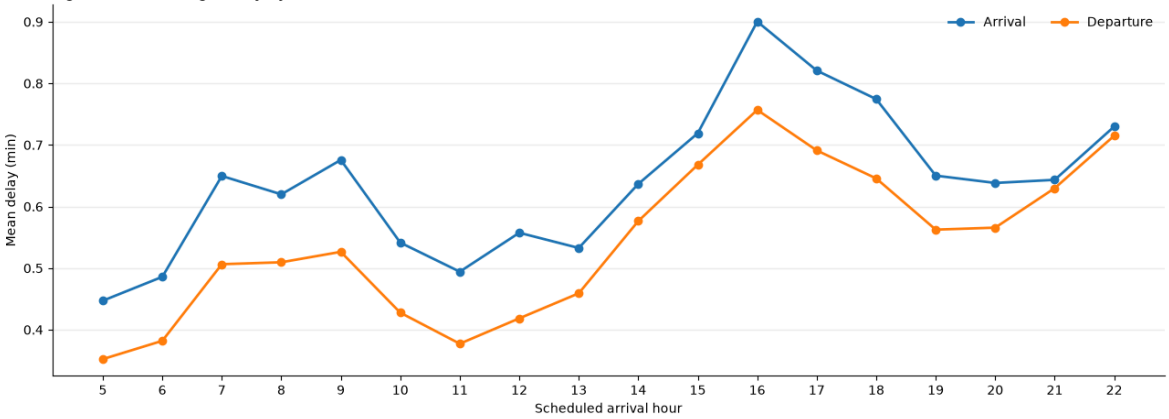


Figure 3.2.3: Frequency of delays

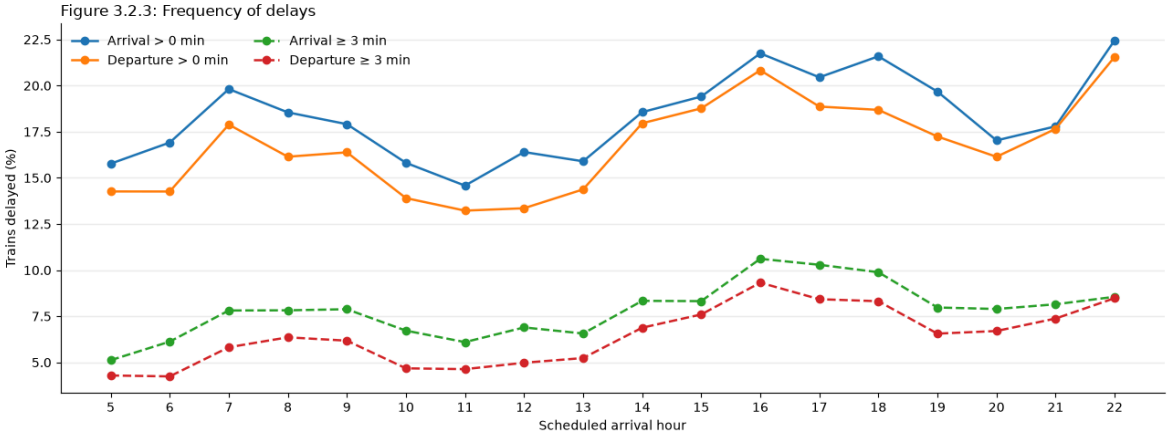


Figure 3.2.4: Cancellation rate by hour

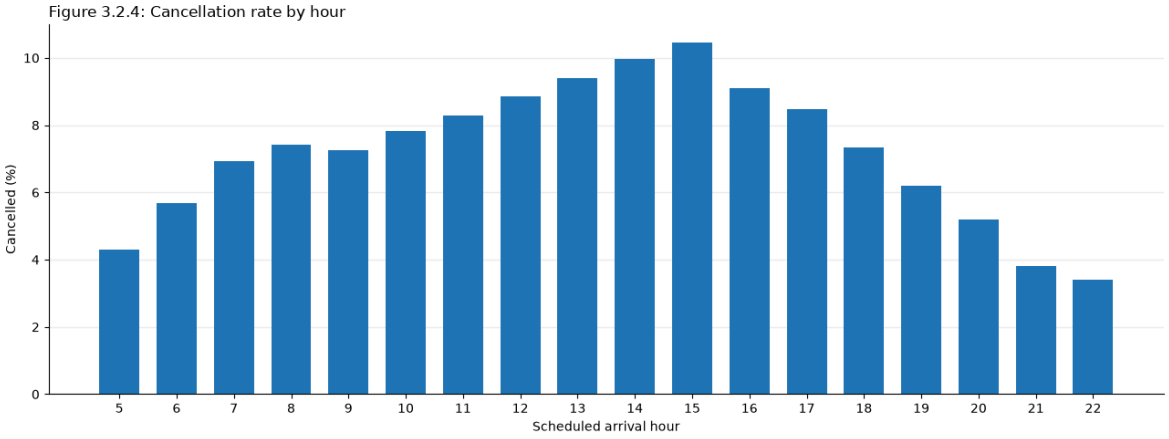


Figure 3.2.5: Average Arrivals and Departures per Hour per Day

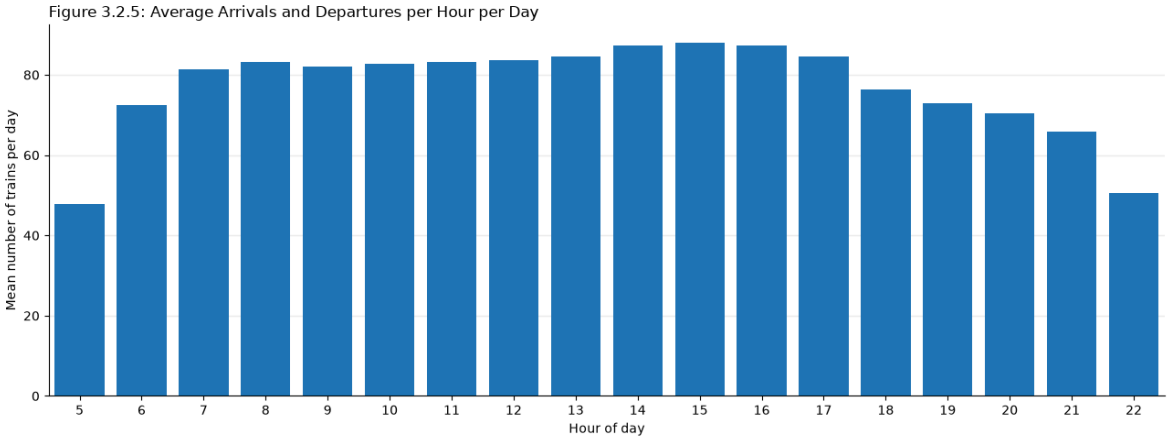


Figure 3.2.6: Average Delayed Arrivals and Departures per Hour (> 0 min)

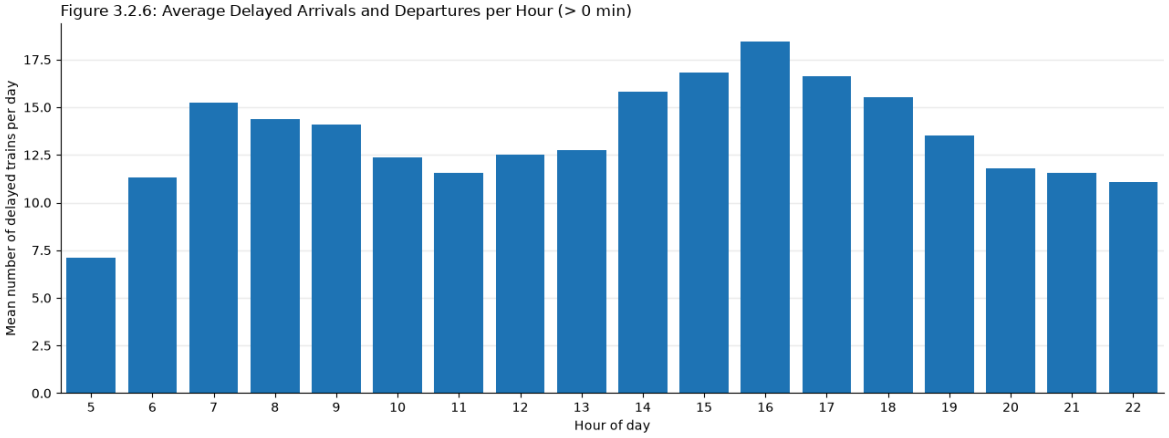


Figure 3.2.2 shows the mean delay in minutes per hour for arrival and departure, the frequency (Fig 3.2.3) of delays in percentage for arrival and departure (> 0 min and >= 3 min), as well as the cancellation rate per hour (Fig 3.2.4).

Delays for a clear daily pattern, where mean arrival delay is lowest in the early morning and around midday (0.45 min and 0.52 min), where it especially rises thorugh the afternoon, peaking in the hour of 16:00 - 17:00 with a mean arrival delay of around 0.9 minutes. An additional peak occurs at 22:00 - 23:00. The share of delayed trains show the same shape: arrivals delayed by more than 0 minutes range from about 15% to 22% in a long afternoon/early evening peak. The afternoon peak coincides with the evening rush, suggesting that the delays build up during the day through knock-on effects and higher loads. Departure delays are consistenly lower than arrivals, in both mean delay and frequency, which again indicate delay absorption during stops using dwell times.

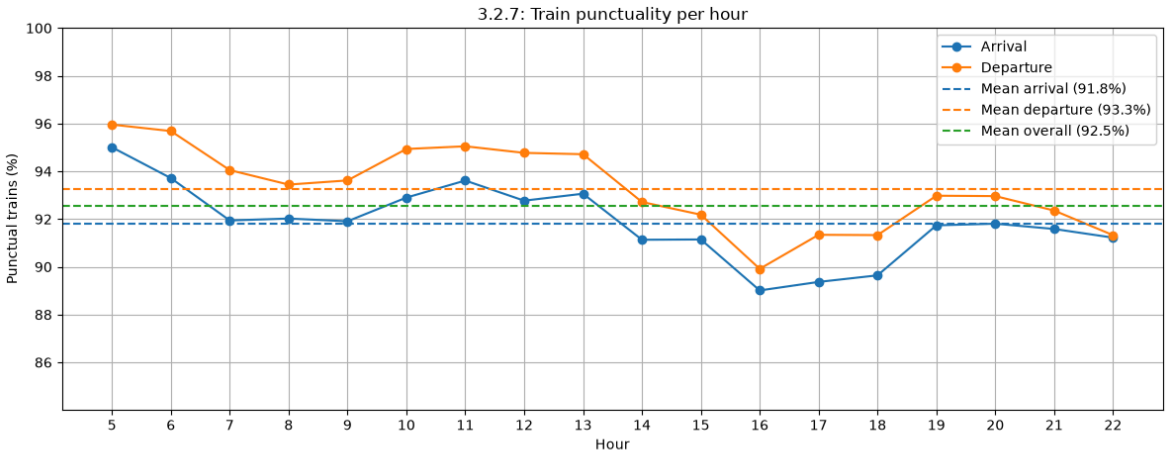
The cancellation rate follows a different pattern; it seems so build up into the afternoon and decreases before the evening rush. Cancellations peak before delay peak hours. A possible explanation is that cancellations are used to prevent propagation or reflect off-peak adjustments, although this cannot be concluded from the figure.

The number of delayed trains are highest in the morning peak and the afternoon peak (Fig 3.2.6).

3.2.3 Punctuality

Punctuality discarded cancellations in the dataset; cancelled trains cannot be punctual.

In [29]:
```
analyse_punctuality(train)
```






```
Trains punctual at both arrival and departure: 90.22%
Trains punctual at arrival: 91.80%
Trains punctual at departure: 93.26%
```

The plot (Fig. 3.2.7 and Fig. 3.2.8) shows train punctuality by hour of day and weekday. Departures are more punctual than arrivals in (nearly) all hours and on all days (mean 93.3% vs 91.8%), indicating possible delay absorption stops.

Punctuality is highest in the early morning between 05:00 and 06:00 and lowest in the afternoon rush-peak between 16:00 and 19:00, with arrival punctuality dropping to around 89%. In the evening, little delay is recovered. Across the week, Tuesday is the least punctual day in the baseline (~89% on arrival and ~91% on departure), whereas Sunday is the most punctual. Weekend days generally perform better than weekdays. The smallest arrival-departure gap is on Friday, and highest on Tuesdays.

At train level, of the almost 9000 trains with a delay at arrival of departure, 54% are delayed at both, 30% are delayed only at arrival and 17% are delayed only at departure, generating delay. Of all delayed arrivals, about 36% are recovered before departure.

Train punctuality of departures is consistently higher than punctuality at arrivals; the mean values confirm this. This is cause for the investigation of delay absorption of Utrecht Centraal.

3.2.4 Delay absorption and propagation

The first plot (Fig. 3.2.9) shows delay recovery or delay increase in per weekday and per hour. The left plots show all delayed trains using a net-delay-change and the right plots show delay absorption of late-arriving trains only.

- In the left panels, blue bars indicate a net delay reduction at the stop and the red bars a net delay increase.
- In the right panels, the blue indicates trains whose delay is absorbed and red trains whose delay propagated.

The net delay change equals the difference between the departure and arrival delay; if net delay change is bigger than zero delays are absorbed or dissolved:

- (departure delay - departure delay) is the actual arrival delay removed.

The delay absorption equals the difference between the departure and arrival delay, divided by the departure delay.

- [(arrival delay - departure delays) / arrival delays] gives the share of arrival delay removed

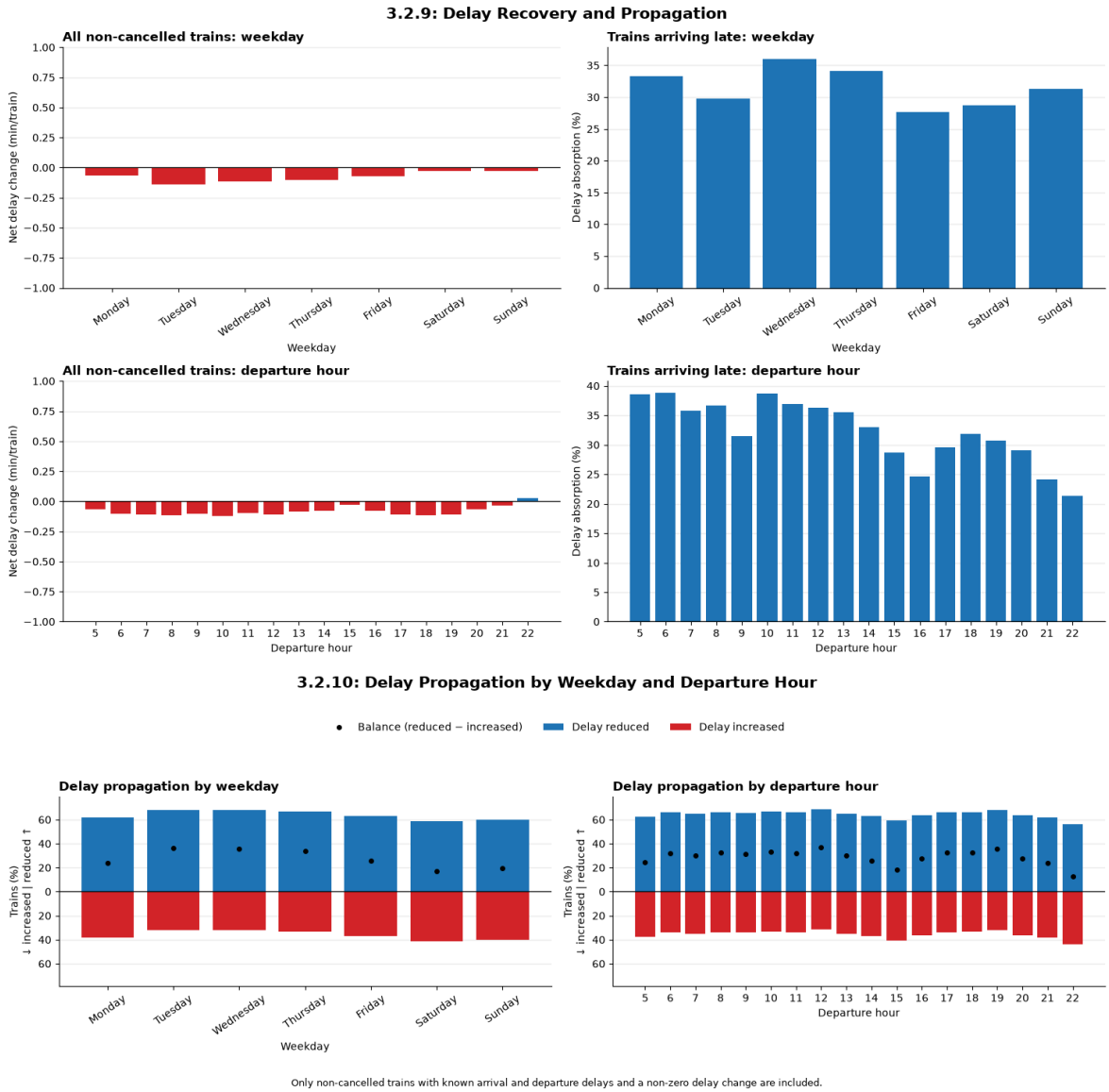
On average, the net delay change is slightly negative (-0.09 min/train), so stops increase delay by a very small, immeasurable, margin. The value per train is small because most trains have no delay to recover. Measured in delay minutes, 12.2% of arrival delay is absorbed, rising to 31.8% for late-arriving trains. From this point on absorption is only calculated for late-arriving trains, as trains arriving on time have no delay to absorb and dilute the statistic. The net delay change is calcualted for all trains, as it also captures delay which is generated by the station.

The plots first column on the statistics of the net delay reduction is an indicative plot showing the minimalistic influence of net delay change on the total dataset; the delay averages down to zero in a sufficiently large dataset.

Absorption is an interesting statistic to show in this stage. Absorption of late trains is highest on Wednesdays (around 36%) and lowest on Fridays (around 27%). By hour, absorption is highest is highest in the early morning and decreases to around 25% in the afternoon. The recovery of capacity is higher in the morning than in the afternoon peak.

In [30]:
```
analyse_delay_absorption(train)
```

```
Mean net delay change (all running trains): -0.09 min/train
Late-arrival delay absorption: 31.76%
```



Only non-cancelled trains with known arrival and departure delays and a non-zero delay change are included.

The bottom plot (Fig. 3.2.10) shows delay change for each weekday and departure hour. Only trains whose delay changes between arrival and departure are included (non-cancelled trains with a known arrival and departure delay, and a non-zero delay change). The blue bars give the share of these trains that leave with less delay than they arrived with; the red bars are the share that leave with more delay. Both shares add up to 100%. The black dots show the balance, which equals the net direction in absorption of propagation.

On every weekday, the majority of trains with a changing delay reduces delay (59-68%). The same trend can be observed on hourly basis.

The bottom plot shows trains, whereas the top figure weigh each train by its delay minutes.

The net delay change and delay absorption show three things:

1. The stop acts on balance as a recovery point: reduction is the most frequent outcome at all days and hours, which is consistent with the positive delay absorption (31.8% of arrivals).

2. Recovery is not complete nor reliable: roughly 1/3th of the trains with a changing delay leave with more delay than they arrived with.
3. The recovery capacity is lowest in the afternoon peak and at the end of the service day, which are also the periods with the lowest punctuality. Indicating these are critical sections in the base selection. Tuesday has the lowest punctuality with the lowest delay absorption in the weekdays.

Use Net delay change (min/train) to answer if the stop, on average, contributes to delay accumulation or absorption (operational effect).

- Use instead of delay absorption as the denominator for arrival delay in absorption can produce misleading or extreme percentages.
- Does not indicate if dwell time improved the delay
- Use when: comparing overall effect of dwell time, compare departure hours or weekdays for all trains, assess overall performance.

Use Delay absorption (%) to answer how effectively existing arrival delay is recovered during the stop.

- Use when: investigate whether late trains recover existing delays, which dwell times help late trains recover

3.2.5 Correlation and dwell-time heatmaps

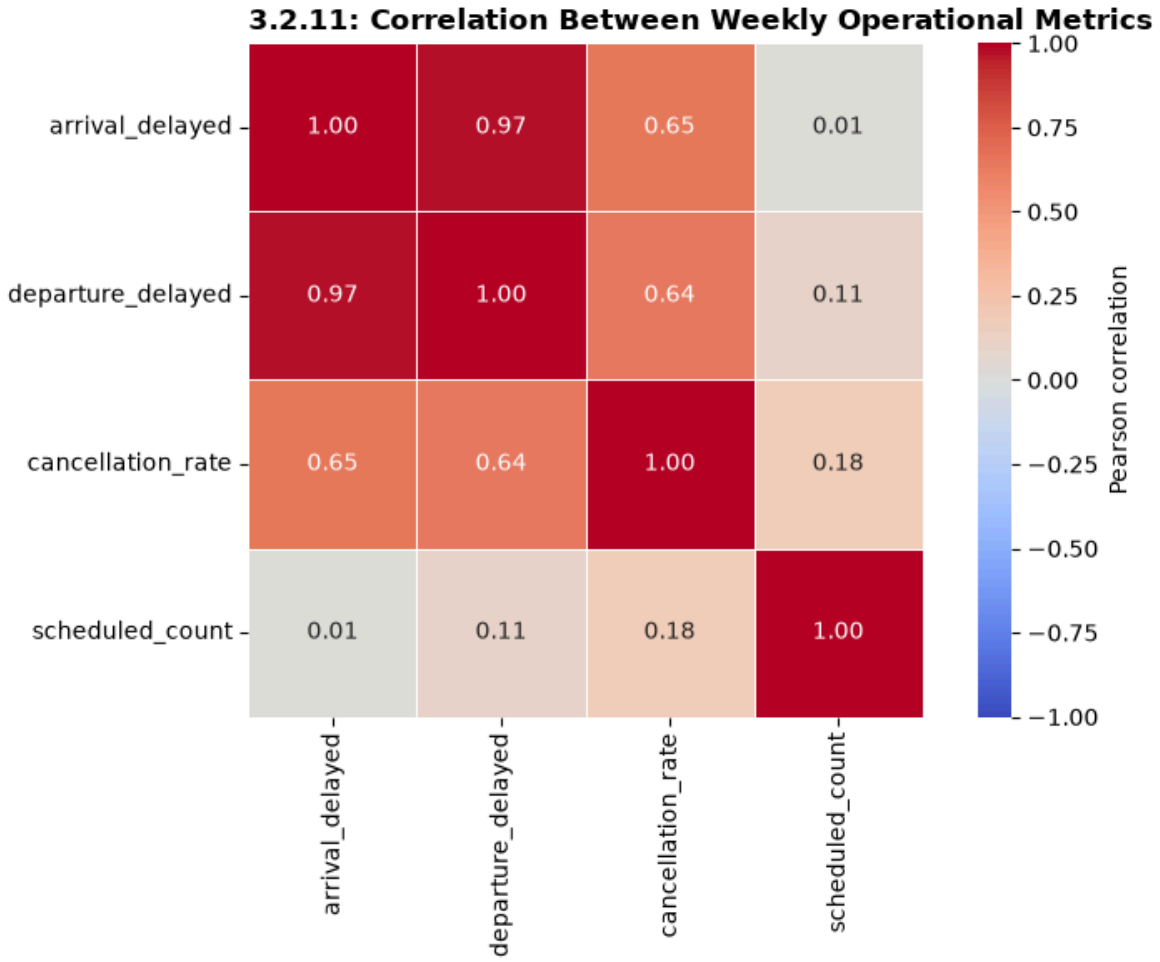
Section 3.2.4 showed that a stop at Utrecht Centraal keeps a net delay equal and reduces delays for late arrivals. Additionally, it showed that this effect shos by weekday and per hour. This section looks at two things; how strongly the weekly operational metrics are related and to what extent the scheduled dwell time at Utrecht Centraal determines or explains the absorption.

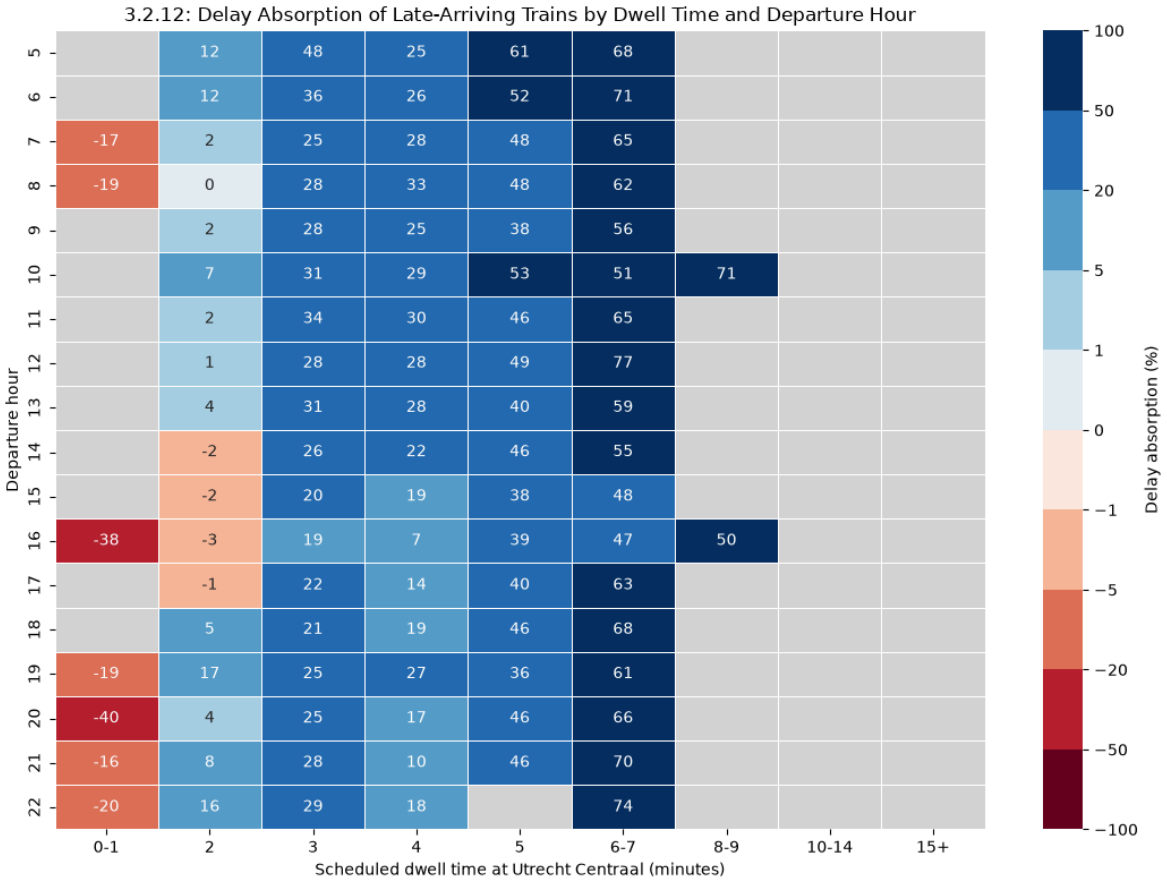
Dwell time determines the available buffer for trains to keep on time.

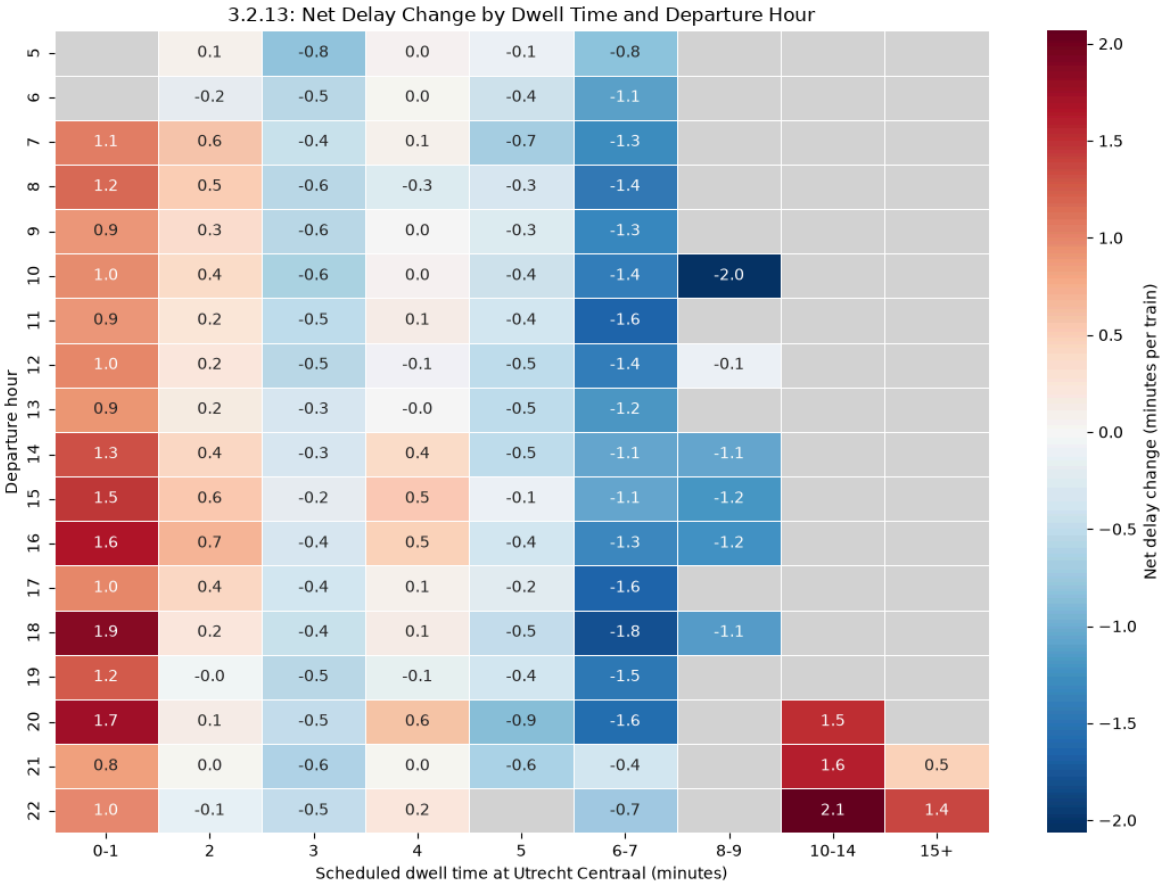
Three heatmaps are used: a weekly correlation matrix, the delay absorption of late-arriving trains by dwell time and departure hour and the mean net delay change of all trains by dwell time and departure hour.

- Heatmaps with less than 15 trains in the corridor are greyed out.

In [31]:
```
analyse_correlations(train)
```


3.2.11: Correlation Between Weekly Operational Metrics
arrival_delayed
departure_delayed
cancellation_rate
scheduled_count
1.00
0.97
0.65
0.01
0.97
1.00
0.64
0.11
0.65
0.64
1.00
0.18
0.01
0.11
0.18
1.00
Pearson correlation
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00


3.2.12: Delay Absorption of Late-Arriving Trains by Dwell Time and Departure Hour
Departure hour
5 12 48 25 61 68
6 12 36 26 52 71
7 -17 2 25 28 48 65
8 -19 0 28 33 48 62
9 2 28 25 38 56
10 7 31 29 53 51 71
11 2 34 30 46 65
12 1 28 28 49 77
13 4 31 28 40 59
14 -2 26 22 46 55
15 -2 20 19 38 48
16 -38 -3 19 7 39 47 50
17 -1 22 14 40 63
18 5 21 19 46 68
19 -19 17 25 27 36 61
20 -40 4 25 17 46 66
21 -16 8 28 10 46 70
22 -20 16 29 18 74
0-1
2
3
4
5
6-7
8-9
10-14
15+
Scheduled dwell time at Utrecht Centraal (minutes)
Delay absorption (%)
100
50
20
5
1
0
−1
−5
−20
−50
−100



1) Weekly correlation (Fig. 3.2.11)

Weekly arrival and departure delays are almost perfectly correlated (r = 0.97), which is almost trivial. Weeks with many late arrivals also have many late departures.

The cancellation rate has a high positive correlation with delays, so poor weeks tend to be poor in arrival and departure delays. The number of scheduled trains is practically unrelated to delays and not related to cancellations (r = 0.01). Disruptive weeks probably drive both delays and cancellations.

2) Delay absorption heatmap (Fig. 3.2.12)

Absorption among late-arriving trains depends mainly on the scheduled dwell time. For a dwell time of 0-1 minute, absorption is negative in all hours with sufficient data (-16% to -40% increase in delay), so these trains leave with more delay than they arrive with. At a scheduled dwell time of 2 minutes the absorption nears zero and steadily rises afterwards from 19% at 3 minutes to 77% at 6-7 minutes.

The departure hour has a second effect. For the same dwell time, the absorption is the highest in the early morning and lowest in the early afternoon; except for scheduled dwell time between 0-1 minutes. The higher afternoon absorption and lower peak-hour absorption (propagation) reflect higher loads on the station and trains in the peak-hours.

3) Net delay change heatmap (Fig. 3.2.13)

The net delay change show the same gradient as in 2) but in minutes per train. For 0-1 minutes of dwell time, net delay change is positive in all hours (+0.8 to +1.9 min), so

delay is generated at these stops. At 2-4 minutes the net change is close to zero and from 4 minutes onwards the net delay change is negative.

The delays turn positive in the afternoon, as in line with the low absorption in the previous heatmap. For long dwell times in the late evening, the net delay is positive; these cells may reflect turnaround or other out of time events and must be treated with caution.

The small average net change (-0.09 min/train) in Section 3.2.4 clearly hides two opposing groups; trains with a short scheduled dwell time which add delay and trains with a longer dwell time which absorb delay.

The following conclusions can be drawn:

1. Delay absorption is mainly determined by the buffer in the timetable. Below 3 minutes of scheduled dwell time, the stop produces delay. Above 5 minutes it removes the largest share of delays.
2. At equal dwell times, the absorption is lower in the afternoon peak. A loaded network thus reduces the effectiveness of the buffer.
3. Weekly delays and cancellations move together.
4. Differences between weekdays and hours in absorption and net change are both dependent on the buffer and network load, not solely on the level of incoming delay.

3.2.6 Baseline conclusions

Over the baseline study period, delays are short and mostly small. The mean arrival delay is within the minute per train, depending on the hour. More than 90% of the trains arrive and depart within 3 minutes of schedule. Of the approximately 9,000 trains with a delay at either arrival or departure, 30% are delayed only at arrival. Cancellations vary strongly throughout the day, but most occur in the afternoon.

A typical day therefore starts with high punctuality and few delays. Delays gradually build up through the day, peak in the afternoon rush period (16:00 - 19:00) and increase again in the late evening. Cancellations peak earlier than delays, already in the early afternoon.

Across the week, Tuesday is the least punctual day and Sunday the most punctual. Across the period, delays are lowest around the turn of the year and highest in mid-February, after which they stabilize. Under normal conditions a delay is small, recurring and has a predictable temporal pattern. Large deviations from this pattern coincide with disruptions and are accompanied by increased cancellations.

The stop of Utrecht Centraal is a recovery point, but only to a limited extent. For late arriving trains, aprroximately one-third of the arrival delay can be removed. This recovery mainly occurs when the timetable provides sufficient scheduled dwell time, particularly at least three minutes, while recovery is more limited during the busy afternoon peak. Trains with a scheduled dwell time of 0-2 minutes systematically accumulate additional delay.

However, differences in delay abosorption and net delay change between hours and weekdays should not be attributed solely to the level of incoming delay. THey are influenced by the interaction between the available buffer time and the operational network load. Periods with greater network pressure, such as the afternoon peak, may therefore provide fewer opportunities to use available dwell time for recovery. The most direct operational lever for improving punctuality at Utrecht Centraal is consequently the provision of sufficient timetable buffer, particularly during periods of high network load.

3.2.7 Disrupted Operations

The study period was identified from the 2nd of January up until the 11th of January, which is displayed in the same way as the previous baseline period in Figures 3.2.7 till 3.2.10.

In [32]:
```
train_selected = load_train_data(start_period='2026-01-02', end_period='2026-01-

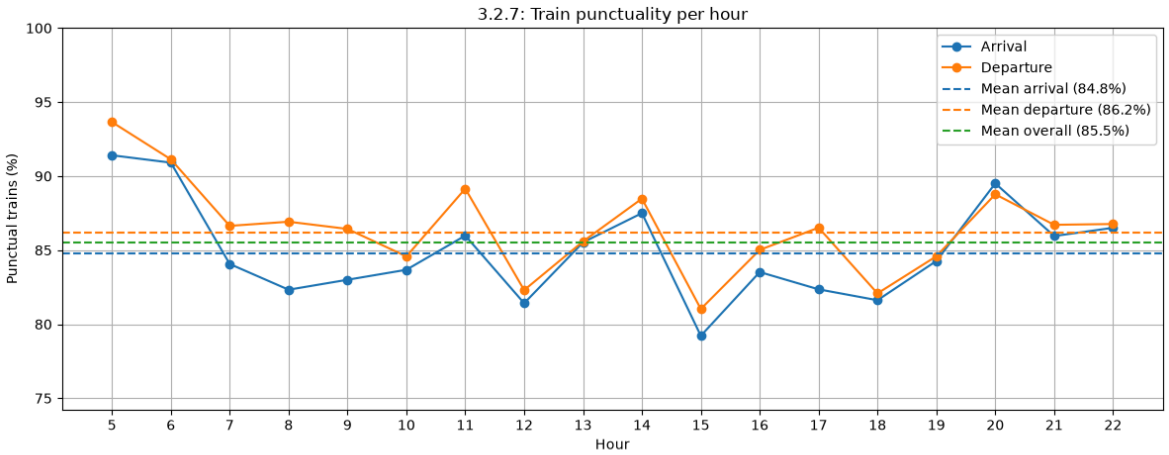
delayed_trains = (len(train_selected[(train_selected['arrival_delayed'] == True)
delayed_operations = len(train_selected[(train_selected['arrival_delayed_3'] ==
total_cancellations = len(train_selected[train_selected['cancelled'] == True])
total_cancellations_perct = total_cancellations / len(train_selected) * 100

print(f"{delayed_trains:.1f}% of trains were delayed.")
print(f"{delayed_operations:.1f}% of train operations were delayed by more than
print(f"{total_cancellations_perct:.1f}% of trains were completely cancelled.")

analyse_delays(train_selected, plot = False)
analyse_punctuality(train_selected)
analyse_delay_absorption(train_selected)
cancellation_chart(train_selected)
```

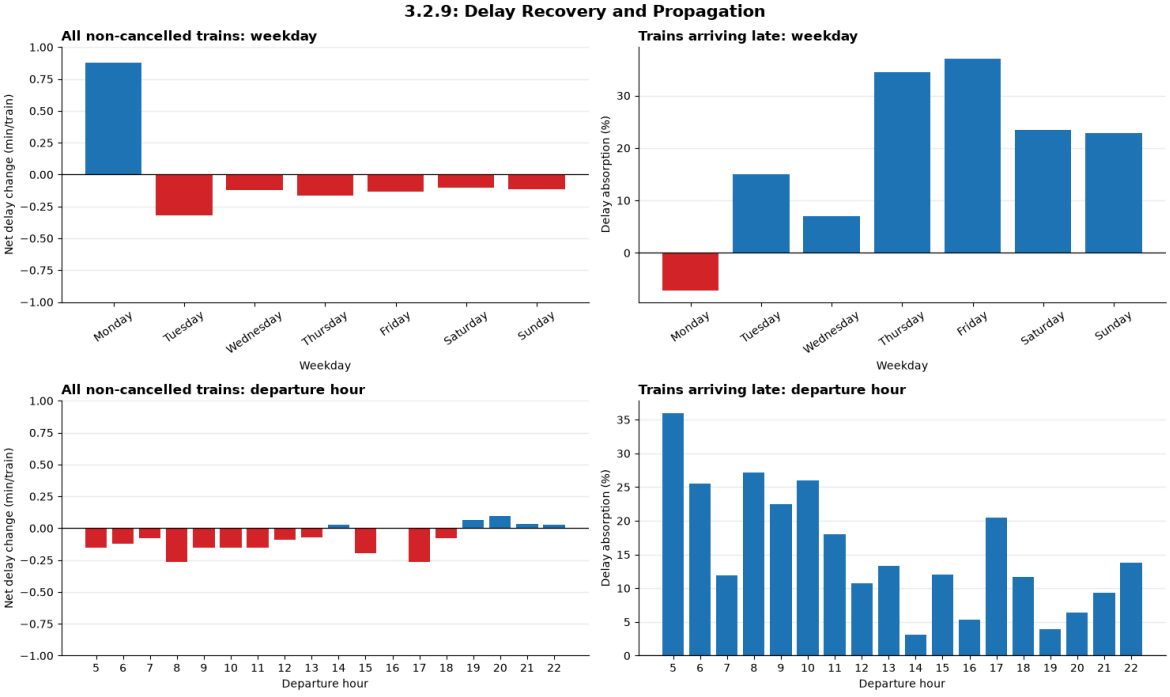
```
26.0% of trains were delayed.
14.0% of train operations were delayed by more than 3 minutes (actual delay class
ification).
44.5% of trains were completely cancelled.
DELAYS AND CANCELLATIONS
Number of complete cancellations: 3456
    26.0% of trains were delayed.
    14.0% of train operations were delayed by at least 3 minutes.
    44.5% of trains were cancelled.

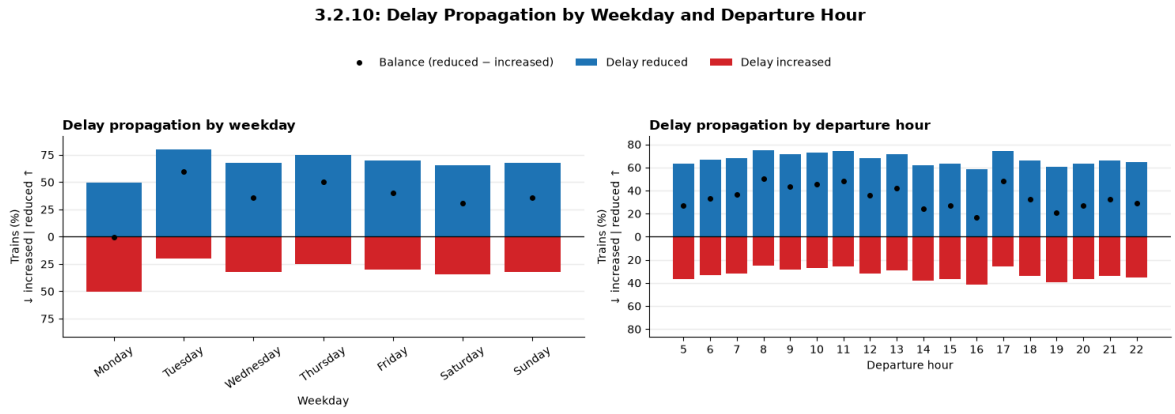
RELATIVE PERFORMANCE
    Delay-minutes per vehicle: 2.22 minutes
    Total delay-minutes per day: 1728.2 minutes
    Cancellations per 100 vehicles: 44.48
    Cancellations per day: 345.60
```

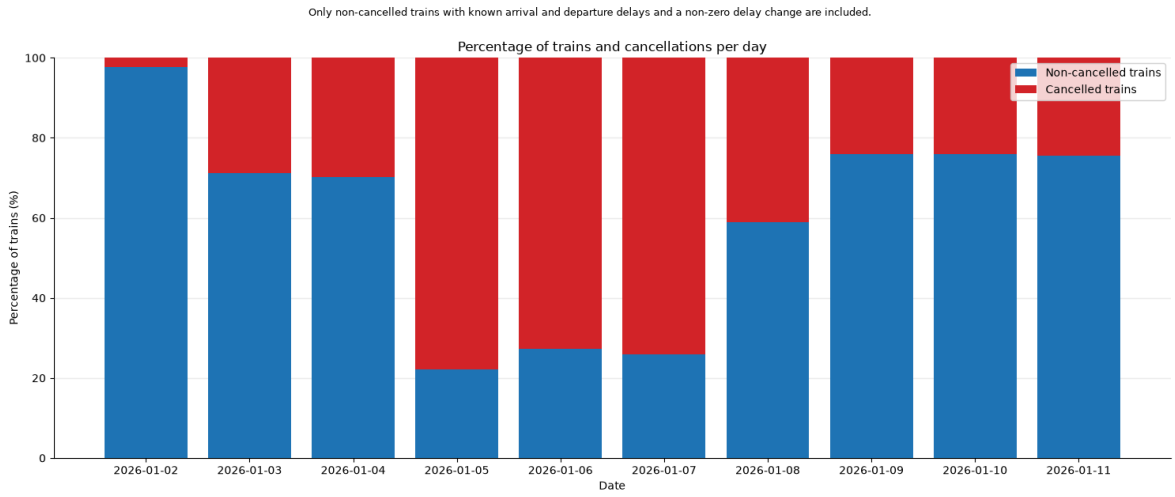




Trains punctual at both arrival and departure: 82.80%
Trains punctual at arrival: 84.77%
Trains punctual at departure: 86.20%
Mean net delay change (all running trains): -0.08 min/train
Late-arrival delay absorption: 13.37%







The indicators of Section 3.2 were recomputed for 2–11 January 2026 without the cancellation filter. 44.5% of scheduled stops were cancelled (3,456 stops; 345.6 per day against 51.9 in the baseline). Of the trains that did run, 26.0% had an arrival or departure delay and 14.0% a delay of at least three minutes (baseline: 24.8% and 9.6%). Mean delay was 2.22 minutes per vehicle (baseline: 1.19). Arrival and departure punctuality were 84.8% and 86.2% (baseline: 91.8% and 93.3%), and 82.8% of trains were punctual at both. Departure punctuality remained above arrival punctuality, so Utrecht Centraal still recovered part of the incoming delay. However, the absorption of late-arriving trains fell to 13.4% (baseline: 31.8%). The mean net delay change was −0.08 min/train, practically equal to the baseline (−0.09). This average is dominated by trains without delay and is therefore insensitive to the disruption.

The impact was concentrated in time. Hourly cancellations exceeded 70% on the afternoon of 4 January and reached almost 100% on the morning of 6 January. Departure punctuality dropped to about 48% on the worst day. Weekday-level values are based on one or two days and should be interpreted as indicative only.

3.3 Comparison analysis

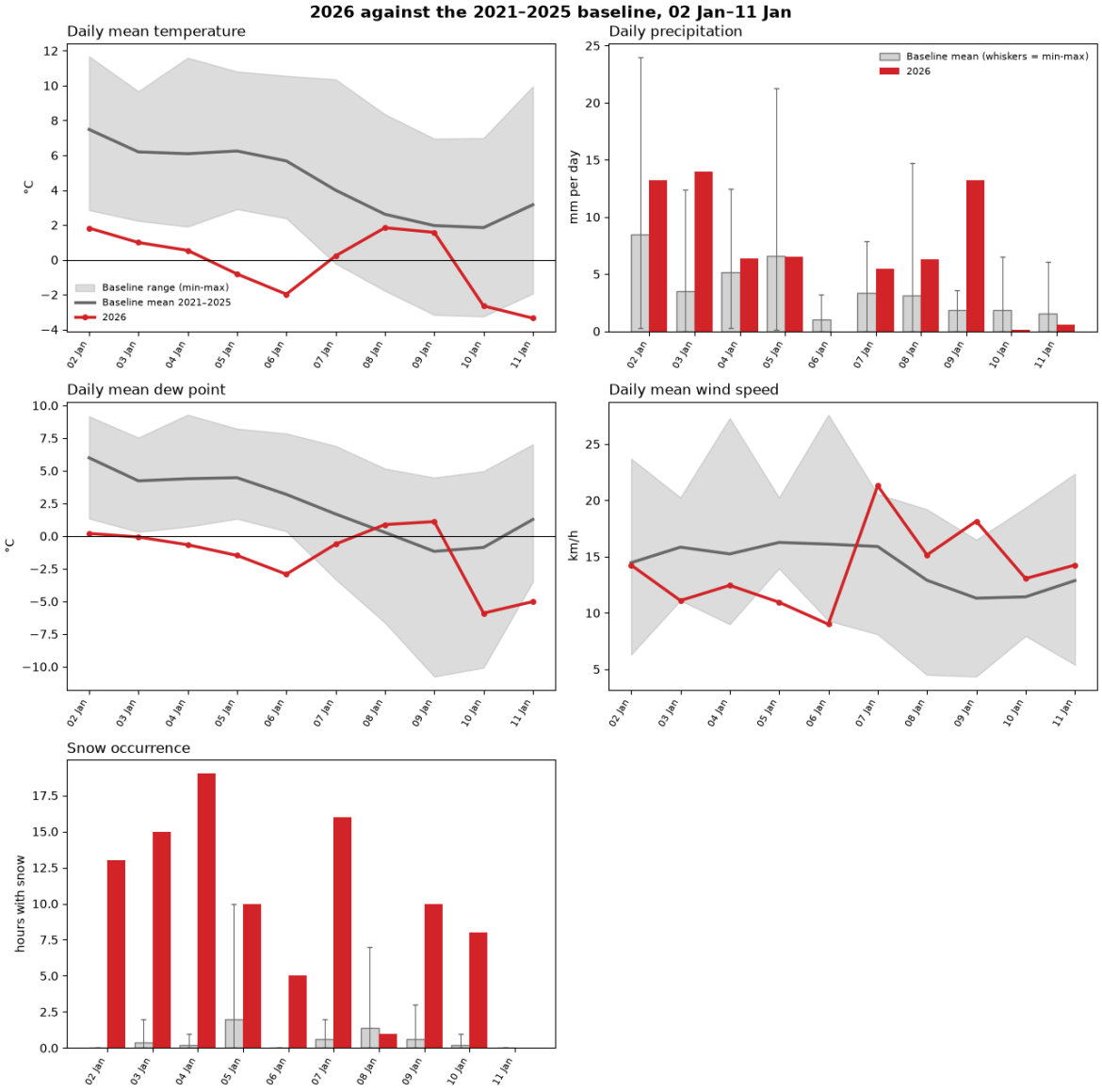
A comparison analysis is carried out between the selected baseline (3.2) and the disrupted operations (3.3), based on the disrupted days identified in Section 3.1. The aim

of this analysis is to determine where, and by how much, operations changed at Utrecht Centraal.

3.3.1 Weather Baseline vs Extreme Weather

To build on the findings and figures in Section 3.1.1, this section added the weather data for the snowstorm event in 2026. The resulting graphs are displayed below. These form the foundation of ascertaining how well the five predetermined weather indicators are able to capture adverse weather conditions, in this particular case the snowstorm event.

In [33]:
```
fig_event_vs_baseline = plot_event_vs_baseline()
```



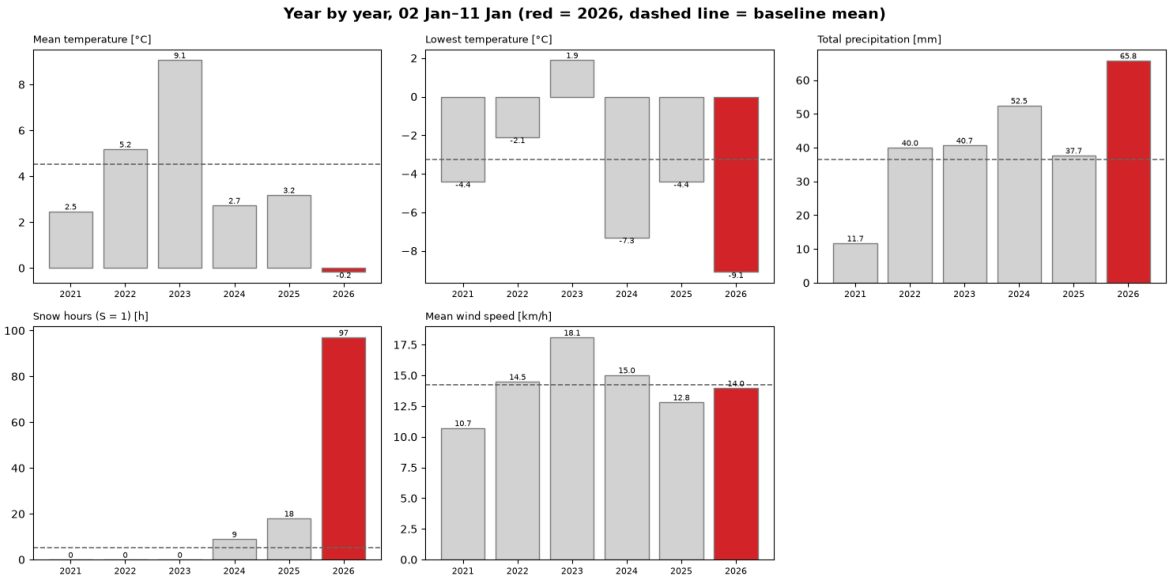
According to preliminary research, the snowstorm occured between January 2nd and January 11th. The following observations regarding the characteristics of the snowstorm can be derived from the above graphs and can be used to test how well the predetermined indicators captured the snowstorm on a day-by-day basis.

- Snow occurence was the strongest indicator, clearly reflecting these adverse weather conditions. It measured outside the typical range on every day, except for January 5th where it reached the range boundary.

- Daily mean temperature and daily mean dew point somewhat reflected the snowstorm in 2026, measuring outside of the typical range on 7 out of 10 days (2-6 January, and on January 11th).
- Daily mean wind speed, on the other hand, lossely captured the snowstorm, performing outside of typical range on 4 out of 10 days (January 5th, 6th, 7th and 9th).
- Daily precipitation was the weakest indicator, poorly capturing the adverse weather conditions, performing outside of the typical range on only 2 out of the 10 days (January 3rd and Janaury 9th).
- Interestingly enough, all five indicators seem to showcase a similar general pattern, decreasing in value from January 2nd up until Janaury 6th, and afterwards (January 7th until January 11th) an upwards trend can be observed. This observation points to January 6th being a turning point in the snowstorm event. In fact, this can potentially explain why Janaury 6th was perceived as the worst day of the snowstorm event (as stated in the introduction).

Besides the daily weather patterns, yearly weather patterns were also analysed to provide a more general view of what typical weather patterns look like in the study period of January 2nd to Janaury 11th. The annual performance of the four of the weather indicators are provided below in their respective graphs, a long with a graph for the lowest temperature in the selected period. This slight change was adopted as a consequence to the previous findings, specifically the similarities shown between daily mean temperature and dew point.

In [34]:
```
fig_yearly = plot_year_by_year()
```



The following observations regarding the characteristics of the snowstorm can be derived from the above graphs and can be used to test how well the predetermined indicators captured the snowstorm on a year-by-year basis.

- Snow occurence still remains the strongest indicator, very clearly performing as an outlier comapred to the previous five years.

- On a similar note, the mean temperature was the lowest in 2026. In fact, it is the only mean below the zero, strongly supporting the claim of adverse conditions in that period. Moreover, the lowest temperature reached in the study period was also the lowest value compared to the previous five years, further supporting the same claim.
- Total precipitation also performed well above the previous years, substantiating the occurence of adverse weather conditions.
- Mean wind speed, however, seems to be the only indicator that did not reflect the adverse weather patterns in 2026, measuring just under to the mean temperature of the previous five years.

Putting it all together, the following key findings have been identified:

- Snow occurrence seems to be the strongest indicator out of the five, clearly showcassing adverse weather patterns on both the daily and yearly level.
- Temperature and dew point also reliabily capture the snowstorm event in 2026. Not only the mean of these indicators reflect this claim, but also the minimum values.
- Total annual precipitation is a strong indicator of adverse weather patterns, properly reflecting for the performance of the snowstorm event. However, on a daily scale, this same metric seemingly fails to adequately capture extreme weather conditions.
- Mean wind speed was consistently a weak indicator of adverse weather conditions.
- Janaury 6th seems to be a relevant turning point in the dataset on a daily scale, potentially explaining why this day was perceived as the worst day of the snowstorm event.

3.3.2 Operations Baseline vs Disrupted Operations

To evaluate the impact of the snowstorm on the operations, the baseline scenario is compared with the snowstorm-disrupted scenario of 2-11 January 2026. The comparison focuses on the previously calculated indicators in Section 3.1 and 3.2. These indicators provide insight in both the continuity of train services and operational performance of the train which remained in service.

In [35]:
```
import background.train_comparison
importlib.reload(background.train_comparison)
from background.train_comparison import compare_operations

table, formatted_table = compare_operations(train, train_selected, baseline_name
     scenario_name='Snowstorm (2-11 Jan)')

display(formatted_table)
```

Figure 3.4.1: Cancellation and Punctuality per Hour Comparison

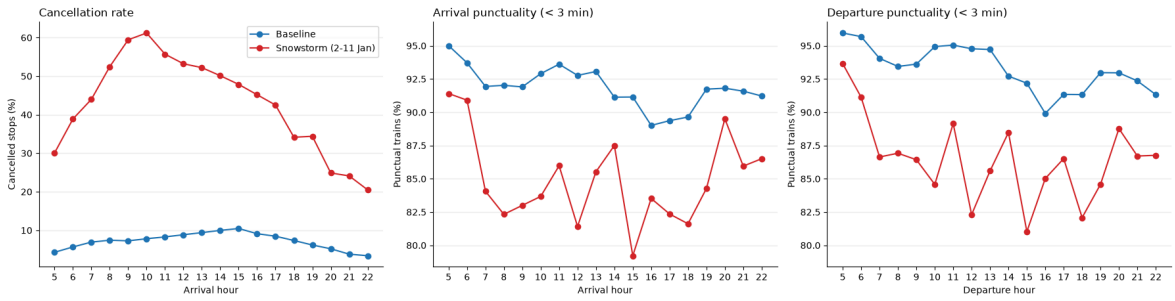


Figure 3.4.2: Cancellation and Punctuality per Weekday Comparison

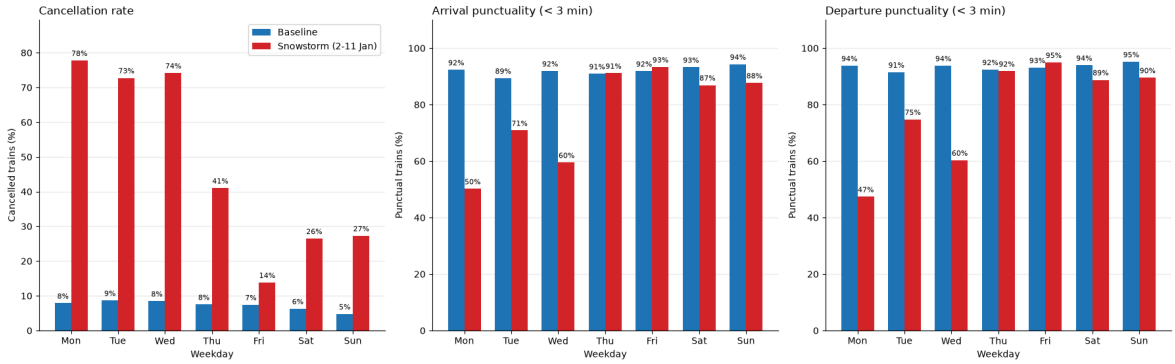
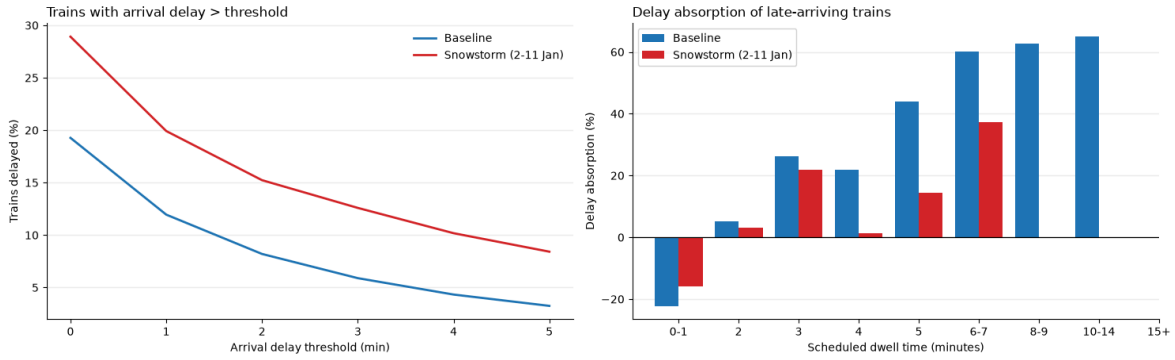


Figure 3.4.2: Delay Size and Delay Absorption, Baseline vs Selected Period



Dwell times with fewer than 15 late-arriving trains or less than 10 arrival delay-minutes are not shown.

Indicator	Baseline	Snowstorm (2-11 Jan)	Difference	Change (%)	Effect
Days analysed (n)	110.0	10.0			
Stops per day	692.3	776.9	84.6	12.2	
Delay minutes per vehicle	1.2	2.2	1.0	87.3	Negative
Cancelled trains (%)	7.5	44.5	37.0	493.6	Negative
Arrival punctuality (%)	91.8	84.8	-7.0	-7.7	Negative
Departure punctuality (%)	93.3	86.2	-7.1	-7.6	Negative
Delay absorption (%)	31.8	13.4	-18.4	-57.9	Negative
Net delay change (minutes)	0.1	0.1	-0.0	-9.6	No effect
Reduced delay (%)	37.8	26.0	-11.8	-31.3	Negative

Figure 3.4.1 shows the cancellation and punctuality per hour compared between the baseline scenario and the snowstorm disrupted scenario. The cancellations are increased with the biggest margin, where on the worst hours cancellations rise from < 10% up until above 60% of the trains cancelled. The arrival and departure punctuality decreased from around 92% to 85% in punctuality, indicating that the trains which were not cancelled, both have lower arrival and departure punctuality in operations.

Figure 3.4.2 displays the same satistics per weekday, for which the same cancellation statistics can be observed. In the arrival and departure punctualities, the most impacted

days can be identified easily in the graph. The departure punctuality reduces from 94% to 47% on monday.

Finally, (Fig 3.4.2) the delay size and absorption have been visualized. In the baseline, <20% of the trains are delayed, whereas in the snowstorm period this is almost equal to 30%. This trend is consistenly higher than the baseline scenario, rather intuitively indicating that the delay and puntuality statistics are in agreement. The delay absorption of late-arriving trains is plotted against the scheduled dwell time in minutes. It can be observed that there is an increased capability of absorbing delays in longer scheduled dwell times in the baseline versus the snowstorm scenarios. However, the constraints may restrict fair results.

The table shows difference of the indicators is in percentage points, whereas change is relative change compared to baseline. The KPI comparison Table confirms a substantial deterioration in operational performance during the snowstorm period. The average number of delay minutes per vehicle increased by 87.3%, while the cancellation rate rose from 7.5% to 44.5% (+37.0 percentage points). Arrival and departure punctuality both decreased by approximately 7 percentage points. Furthermore, delay absorption decreased from 31.8% to 13.4%, indicating a reduced ability to recover delays within the remaining train operations. Overall, the results show that the snowstorm affected both the continuity of train services and the performance of trains that remained in operation.

These indicators are conditional on running trains and therefore understate the impact. Fewer trains operate during disruptions, and when NS and ProRail proactively reduce capacity, the remaining services can perform better. Combining cancellations and punctuality shows the effect: the share of scheduled trains that ran punctually fell from about 83.5% to about 46.0%. Performance indicators should therefore always be read together with the cancellation rate.

3.3.3 Extreme Weather vs Disrupted Operations

To further analyse the performance of the trainnetwork during the snowstorm the performance indicators are measured against the weather perfomance indicators during the period of the snowstorm. The researched period is longer than the period from the previous chapters to have a better view at what the performance was before and after the storm.

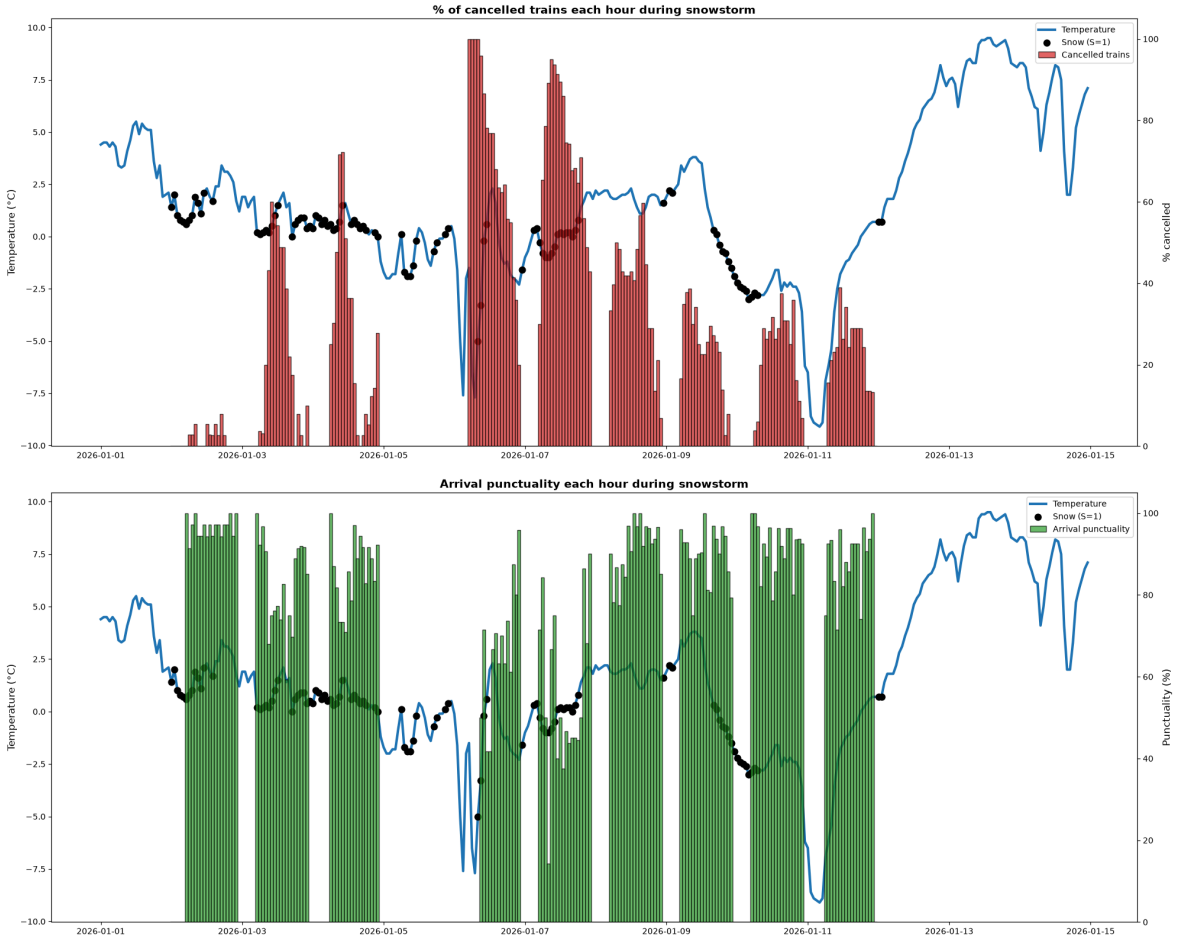
The following train performance indicators are looked at:

- % of cancellations
- Punctuality

In [36]:
```
from background.weather_comparison import (
    analyse_delays_per_hour,
    cancellation_chart,
    cancellation_snow_chart,
    punctuality_chart
)
```

```
df_train = load_train_data(
    start_period='2026-01-02',
    end_period='2026-01-11'
)

cancellation_snow_chart(df_train, df_weather)
punctuality_chart(df_train, df_weather)
```



From Figure 3.5 it can be observed that the cancellations and therefore the disruptions start on the 3rd of January with the highest peaks on the 5th, 6th and 7th. It is worth noting that there are few disruptions during the first snowfalls during the 2nd. At the high peaks there are even some hours when the cancellation rate is 100%. The disruptions continue through the whole week period and stabilize to a decent performance at the end of the period. At the end of the period the temperature rises to about 10°C and the snow is likely melted.

Figure 3.6 shows that also the punctuality takes a hit from the 5th until the 7th. After these days the punctuality recovers a bit but it is still not very high compared to the baseline punctuality of around 91.8%. So the performance is still not at a normal level.

Discussion & Limitations

Discussion

The snowstorm primarily affected the continuity of services at Utrecht Centraal and to a lesser extent the performance of the trains which continued operation. This suggests that the main impact was a reduction in the operating trains, rather than substantially higher delays than those still running. This can be cause for further analysis and possible indicator/criteria linking delay-minutes to cancellations directly. The data does not distinguish why a train has been cancelled (infrastructure failure or proactive timetabling), so their contributions are unknown. Investigating these specific causes in future research could provide deeper insight into service disruptions and help train operators develop targeted solutions.

Delay absoprtion among late-arriving trains at Utrecht Centraal decreased substantially, indicating a decreased effectiveness in recovery during the storm. Either reduced operational buffer, increased pressure on the system or changes in the service composition could explain this decline. This is cause for further investigation on a more microscopic level. The high cancellation rate highlights that the performance indicators (net delay change, delay absorption and punctuality) do not tell the full story and are strongly influenced by the number of trains that remain in operation. When anticipating adverse weather conditions, ProRail and NS may proactively reduce available capacity or adjust the timetable to maintain more manageable operations. As a result, fewer trains operate on the network, potentially improving the performance indicators of the remaining services.

The establishment of the baseline for weather data requires careful consideration, as even a five-year dataset displays considerable variation, and thus quite a broad range of what is deemed a typical weather pattern. Furthermore, because this research relied on a single weather station, it carries a higher risk of sensor malfunctions, such as incorrect or missing data, a vulnerability that can reasonably intensify during extreme weather events. To mitigate this risk, future studies should aggregate data from multiple sources, such as additional weather stations surrounding Utrecht Centraal. Finally, a broader range of weather indicators should be explored, as the four reliable indicators used in this research only captured a limited dimension of local weather patterns.

Limitations

During the snowstorm, there was also an IT outage that may have further increased the disruptions during this period. The impact of this event is not analysed in this report.

Additionally, the relationship between weather and operational performance is only based on a 10-day descriptive comparison instead of a (prolonged) quantitative analysis with statistical testing. Therefore, no causal conlusions can be drawn regarding the actual extent to which the observed disruptions were caused directly by the weather conditions.

Conclusion & Recommendations

This report investigated the performance of the train network at Utrecht Centraal during the snowstorm from 2 to 11 January 2026, compared with a typical period.

When observing weather conditions, snow occurence was the strongest indicator of the storm on both a daily and yearly reoccurance level. Tempearture and dew point also reflected the changing conditions, but to a lesser extent. The precitipation was weak on a daily level, but a strong indicator on a yearly level. The wind speed was a weak indicator. The morning of the 6th of January combined sub-zero temperatures with temperatures equal to the dew point, marking a turning point in the weather data. The disruption coincided with the conditions following the precipitation, instead of the snowfall itself.

Comparing the baseline train performance to the performance during the snowstorm shows the following results: the cancellation rate rose from 7.5% to 44.5% (+37%), with a near-complete cancellation of services in the morning of 6 January. For trains that did run, mean delays per vehicle increased by 87%, while arrival and departure punctuality statistics fell by around 7% in total. Furthermore, delay absorption of late-arriving trains fell from 31.8% to 13.4%, which is a usefull function of the hub-function of Utrecht Centraal.In normal operations, Utrecht Centraal functions as a (limited) recovery point, where absorption depends on the scheduled dwell time, stops under three minutes tend to generate or keep delay. Considering all events together, the snowstorm caused the share of scheduled trains running on time to decrease from 83.5% to around 46%. The main impact was therefore on the service continuity, with a smaller deterioration in the quality of the remaining service.

In conclusion, during the January 2026 storm both the continuity and the quality of service at Utrecht Centraal substaintially decreased, with cancellations being the main event.

Recommendations

Recommendations can be created for NS and ProRail in specific, as well as for further research and analysis.

NS and ProRail should consider scheduling a minimum dwell time of three minutes where operationally feasible, particularly during afternoon peak hours and in expected heavy-winter timetables. In the baseline, stops with a scheduled dwell time of only 0-2 minutes systematically generated or retained their current delay. Although this is trivial, dwell times of three minutes already absorbed more of the delays. During the storm the absorption of late-arriving trains showed that the buffer was the least effective when it was needed the most. However, longer dwell times consume more track- and platform capacities and is dependent on train composition. Increasing dwell times could improve resilience. The effect found is not causal and cause for further timetabling research. A second recommendation would be to plan the recovery of operations after the event, as punctuality remained below the baseline after the peak disrupted days. A return to service should be faster than days after a disrupted event, this can be achieved using better disruption prediction or rescheduling strategies.

An important first step for future research is to establish a train performance baseline consistent with the weather baseline. The current comparison is affected by differences in season and historical timetables, which limits the comparability of the results. However, this tests robustness. The robustness of the findings should be evaluated by repeating the analysis on other stations and affected areas, helping to determine whether the observed reduction in performance patterns occur elsewhere in the network. The relation between weather and operations should be quantified instead of described. Statistical methods such as regression or machine learnign techniques, could identify and quantify the influence of weather variables on performance. These findings could contribute to developing a forecast-based decision-support tool for NS and ProRail. Such tools could use (predicted) weather conditions and their operational impact as triggers for timetable adjustments and disruption management strategies.

AI Statement:

We, as a group, acknowledge the use of ChatGPT, Gemini and Claude in preparing this submission. We used it for the following purposes:

- Brainstorming ahead of team meetings
- Concatenating data flows after creation (e.g. having AI assist to create a code which was in a .ipynb to convert to a .py after validation to use as an import definition)
    - In specific, train_analysis.py was created using chatGPT where period_selection.ipynb was its input with the initial prompt to create runnable definitions per sub-section, later ammended manually for additions without AI usage.
- Providing feedback/code on how to improve basic matplotlib plots
- Cleaning datasets after convertion from text file to CSV file (e.g. removing unnecessary blank spaces)
- Helping in solving errors occuring during coding

We confirm that we have critically reviewed and verfied all AI-assisted content per part used and that the submitted work reflects our own understanding. We take full responsibility for the accuracy and orginality of the combination of AI usage and project applicance.

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